Guide to the WMO Information System (WMO-No. 1061), Volume II - WMO Information System 2.0 (APPROVED)

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Standing Committee on Information Management and Technology (SC-IMT)[1]

Commission for Observation, Infrastructure and Information Systems (INFCOM)[2]

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Purpose of this Guide

In conjunction with the *Manual on the WMO Information System* (WMO-No. 1060), Volume II – WMO Information System 2.0 (*Manual on WIS*, Volume II), the present *Guide to the WMO Information System* (WMO-No. 1061), Volume II – WMO Information System 2.0 *(Guide to WIS*, Volume II) is designed to ensure adequate uniformity and standardization in the data, information and communication practices, procedures and specifications employed by WMO Members in the operation of the WMO Information System 2.0 (WIS2) as it supports the mission of the Organization. The *Manual on WIS*, Volume II contains standard and recommended practices, procedures and specifications. The *Guide to WIS*, Volume II contains additional information concerning practices, procedures and specifications that Members are invited to follow or implement in establishing and conducting their arrangements in compliance with the WMO technical regulations and in developing meteorological and hydrological services.

[1] https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt

[2] https://community.wmo.int/governance/commission-membership/infcom

PART I. Introduction

1.1 Background

Since the Global Telecommunication System (GTS) entered operational life in 1971, it has been a reliable real-time exchange mechanism of essential data for WMO Members.

In 2007, the WMO Information System (WIS) entered operation to complement GTS, providing a searchable catalogue and a Global Cache to enable additional discovery, access and retrieval of data. The success of WIS was limited, as the system only partially met the requirement of providing simple access to WMO data. Today's technology developed for the Internet of Things (IoT) opens the possibility of creating a WIS2 that is able to deliver an increasing number and volume of real-time data to WMO centres in a reliable and cost -effective way.

WIS2 has been designed to meet the shortfalls of the current WIS and GTS, support Resolution 1 (Cg-Ext(2021)) – WMO Unified Policy for the International Exchange of Earth System Data (*World Meteorological Congress: Abridged Final Report of the Extraordinary Session* (WMO-No. 1281)), support the Global Basic Observing Network (GBON) and meet the demand for high data volume, variety, velocity and veracity.

The WIS2 technical framework is based around three foundational pillars: leveraging open standards, simpler data exchange and cloud-ready solutions.

1.1.1 Leveraging open standards

WIS2 leverages open standards to take advantage of the ecosystem of technologies available on the market, thereby avoiding the need to build bespoke solutions that can force National Meteorological and Hydrological Services (NMHSs) to procure costly systems and equipment. In today's standards development ecosystem, standards bodies work together closely to minimize overlap and build on one another's areas of expertise. For example, the World Wide Web Consortium provides the framework of web standards, which the Open Geospatial Consortium (OGC) and other standards bodies leverage. WIS2 leverages open standards with industry adoption and wider, stable and robust implementations, thus extending the reach of WMO data sharing and lowering the barrier to access by Members.

1.1.2 Simpler data exchange

WIS2 prioritizes public telecommunication networks, rather than private networks, for GTS links. The Internet is therefore the best choice for a local connection, as it uses commonly available and well-understood technology.

WIS2 aims to improve the discovery, access and utilization of weather, climate and water data by adopting web technologies proven to provide a truly collaborative platform for a more participatory approach. Data exchange using the Web also facilitates easy access mechanisms. Browsers and search engines allow web users to discover data without the need for specialized software. The Web also enables additional data access platforms, such as desktop geographical information systems (GIS), mobile applications and forecaster workstations. The Web provides access control and security mechanisms that can be utilized to freely share core data as defined by

Resolution 1 (Cg-Ext(2021)) and to protect the data with more restrictive licensing constraints. Web technologies also allow for authentication and authorization to enable the provider to retain control of who can access published resources and to request users to accept a license specifying the terms and conditions for using the data as a condition of being granted access.

WIS2 uses a "publish-subscribe" pattern by which users subscribe to a topic to receive new data in real time. The mechanism is similar to WhatsApp and other messaging applications. It is a reliable and straightforward way to allow users to choose their data of interest and to receive them reliably.

1.1.3 Cloud-ready solutions

The cloud provides reliable platforms for data sharing and processing. It reduces the need for expensive local IT infrastructure, which constitutes a barrier to developing effective and reliable data processing workflows for some WMO Members. WIS2 encourages WMO centres to adopt cloud technologies, where appropriate, to meet users' needs. While WMO technical regulations will not mandate cloud services, WIS2 will promote the gradual adoption of cloud technologies that provide the most effective solution.

The cloud-based infrastructure allows for the easy portability of technical solutions, ensuring that a system implemented by a specific country or territory can be packaged and deployed easily in other countries/territories with similar needs. In addition, using cloud technologies allows WIS2 to deploy infrastructure and systems efficiently, while requiring minimal effort from the NMHSs, by shipping ready-made services and implementing consistent data processing and exchange techniques.

It is important to note that hosting data and services on the cloud does not affect data ownership. Even in a cloud environment, organizations retain ownership of their data, software, configuration and change management as if they were hosting their infrastructure. As a result, data authority and provenance stay with the organization, and the cloud is simply a technical means to publish the data.

1.1.4 Why are datasets so important?

WMO enables the international exchange of observations and model data for all Earth-system disciplines.

Resolution 1 (Cg-Ext(2021)) describes the Earth system data that are necessary for efforts to monitor, understand and predict the weather and climate – including the hydrological cycle, the atmospheric environment and space weather.

WIS2 is the mechanism by which these Earth system data are exchanged.

A common practice when working with data is to group them into datasets. All the data in a dataset share some common characteristics. The Data Catalog Vocabulary (DCAT) defines a dataset as a "collection of data, published or curated by a single agent, and available for access or download in one or more representations".[1]

Why is this important? The "single agent" (such as an organization) responsible for managing the collection ensures consistency among the data. For example, in a dataset:

- All the data should be of the same type (for example, observations from weather stations);
- All the data should have the same license and/or usage conditions;
- All the data should be subject to the same quality management regime - which may mean that all the data are collected or created using the same processes;
- All the data should be encoded in the same way (for example, using the same data formats and vocabularies);
- All the data should be accessible using the same protocols - ideally from a single location.

This consistency means that it is possible to predict the contents of a dataset, at least regarding the common characteristics, making it easier to write applications to process the data.

A dataset may be published as an immutable resource (such as data collected from a research programme), or it may be routinely updated (for example, every minute, as new observations are collected from weather stations).

A dataset may be represented as a single, structured file or object (for example, a CSV file in which each row represents a data record) or as thousands of consistent files (for example, output from a reanalysis model encoded as many thousands of General Regularly-distributed Information in Binary form (GRIB) files). Determining the best way to represent a dataset is beyond the scope of this Guide – there are many factors to consider. The key point here is that the dataset is considered a single, identifiable resource irrespective of how it is represented.

Because data are grouped into a single, conceptual resource (that is, the dataset) it is possible to:

- Assign this resource an identifier and use this identifier to unambiguously refer to collections of data;
- Make statements about the dataset (that is, metadata) and infer that these statements apply to the entire collection.

The dataset concept is central to WIS2:

- Discovery metadata about datasets are published, as specified in the *Manual on WIS*, Volume II – Appendix F. WMO Core Metadata Profile (Version 2);
- Data consumers can search for datasets that contain relevant data using the Global Discovery Catalogue (see 2.4.4 Global Discovery Catalogue);
- Data consumers can subscribe to notifications about updates to a dataset via a Global Broker (see 2.4.2 Global Broker);
- Data consumers can access the data that comprise a dataset from a single location using a well-described mechanism.

It is up to data publishers to decide how their data are grouped into datasets – effectively, to decide what datasets they publish to WIS2. That said, it is recommended that, subject to the consistency rules above, data publishers should organize their data into as few datasets as possible.

For a data publisher, this means fewer discover metadata records to maintain. For a data consumer, this means fewer topics to subscribe to and fewer places to access the data.

There are some things that are fixed requirements for datasets:

1. All data in the dataset must be accessible from a single location;
2. All data in the dataset must be subject to the same license or usage conditions.

Some examples of datasets include:

- The most recent five days of synoptic observations for an entire country or territory; [2]
- A long-term record of observed water quality for a managed set of hydrological stations;
- The output from the most recent 24 hours of operational numerical weather prediction model runs;
- The output from six months of experimental model runs. It is important to note that the output from operational and experimental model runs should not be merged into the same dataset because they use different algorithms - it is very useful to be able to distinguish the provenance (or lineage) of data;
- A multi-petabyte global reanalysis spanning 1950 to the present.

In summary, datasets are important because they are how data are managed in WIS2.

1.2 Information for the data consumer

Data consumers wanting to use data published via WIS2 should read the guidance presented here. In addition, a list of references to informative material in this Guide and elsewhere is provided at the end of this section.

1.2.1 How to search the Global Discovery Catalogue to find datasets

The first step to using data published via WIS2 is to determine which dataset or datasets contain the data that are needed. To do this, a data consumer may browse discovery metadata provided by the Global Discovery Catalogue. Discovery metadata follow a standard scheme (see *Manual on WIS*, Volume II – Appendix F. WMO Core Metadata Profile (Version 2)). A data consumer may discover a dataset using keywords, a geographic area of interest, temporal information, or free text. Matching search results from the Global Discovery Catalogue provide high-level information (title, description, keywords, spatiotemporal extents, data policy, licensing, contact information), from which data consumers can assess and evaluate their interest in accessing/downloading data associated with the dataset record.

A key component of dataset records in the Global Discovery Catalogue is "actionable" links. A dataset record provides one or more links, each clearly identifying its nature and purpose (informational, direct download, application programming interface (API), subscription) so that the data consumer can interact with the data accordingly. For example, a dataset record may include a link to subscribe to notifications about the data(see 1.2.2 How to subscribe to notifications about the availability of new data), or an API, or an offline archive retrieval service.

The Global Discovery Catalogue is accessible via an API and provides a low-barrier mechanism (see 2.4.4 Global Discovery Catalogue). Internet search engines are able to index the discovery metadata in the Global Discovery Catalogue, thereby providing data consumers with an alternative means to search for WIS2 data.

1.2.2 How to subscribe to notifications about the availability of new data

WIS2 provides notifications about updates to datasets, for example, a notification may indicate that a new observation record from an automatic weather station has been added to a dataset of surface observations. These notifications are published on Message Brokers. Where data consumers need to use data rapidly once they have been published (for example, as inputs to a weather prediction model), they should subscribe to one or more Global Brokers to get notification messages using Message Queuing Telemetry Transport (MQTT) protocol.[3]

In WIS2, notifications are republished by Global Brokers to ensure resilient distribution. Consequently, there will be multiple places where one can subscribe. Data consumers requiring real-time notifications must subscribe to Global Brokers. Data consumers should subscribe to more than one Global Broker to ensure that notifications continue to be received if a Global Broker instance fails.

A dataset in WIS2 is associated with a unique topic. Notifications about updates to a dataset are published to the associated topic. Topics are organized according to a standard scheme (see the *Manual on WIS*, Volume II - Appendix D. WIS2 Topic Hierarchy).

A data consumer can find the appropriate topic to subscribe to either by searching the Global Discovery Catalogue, by using an Internet search engine,[4], or by browsing the topic hierarchy on a Message Broker.

WIS2 uses Global Caches to distribute core data, as defined in the WMO Unified Data Policy (Resolution 1 (Cg-Ext (2021))). Each Global Cache republishes core data on its own highly available data server and publishes a new notification message advertising the availability of those data from the Global Cache location.

Notifications from WIS2 Nodes and Global Caches are published on different topics: The root topic used by WIS2 Nodes is `origin`, while the root topic used by Global Caches is `cache`. Other than the root, the topic hierarchy is identical. For example, for synoptic weather observations published by Environment Canada:

- Environment and Climate Change Canada, Meteorological Service of Canada's WIS2 Node, publishes to: `origin/a/wis2/ca-eccc-msc/data/core/weather/surface-based-observations/synop`;
- Global Caches publish to: `cache/a/wis2/ca-eccc-msc/data/core/weather/surface-based-observations/synop`.

As per clause 3.2.13 of the *Manual on WIS*, Volume II, data consumers should access core data from the Global Caches. In order to access these data, they must subscribe to the `cache` topic hierarchy. They will then receive the relevant notifications from the Global Caches, each of which will contain a link (URL) enabling them to download the relevant data from the data server of the corresponding Global Cache.

1.2.3 How to use a notification message to decide whether to download data

On receipt of a notification message, a data consumer needs to decide whether to download the newly available data. The content of the notification message provides the information needed to make this decision (see the *Manual on WIS*, Volume II - Appendix E. WIS2 Notification Message).

In many cases, data consumers will use a software application to determine whether or not to download the data. The present section explains this process.

When subscribing to multiple Global Brokers, data consumers will receive multiple copies of a notification message. Each notification message has a unique identifier, defined using the `id` property. Duplicate messages should be discarded.

Core data are available from both a WIS2 Node and the Global Caches, each of which will publish a different notification message advertising an alternative location from which the data may be downloaded. Because these are different messages, they will have different identifiers. However, each of these messages refers to the same data object, which is uniquely identified in the notification message using the data_id property. Notification messages from different sources can easily be compared to determine whether they refer to the same data. By subscribing to the cache root topic, data consumers will only receive notifications about data available from the Global Caches. The origin root topic should be used when subscribing to notifications about recommended data. Data consumers should not subscribe to the origin root topic for notifications about core data because the notification messages provided on these topics will refer to data published directly on the WIS2 Nodes (referred to as the "origin").

Data consumers need to consider their strategy for managing these duplicate messages. From a data perspective, it does not matter which Global Cache instance is used – they will all provide an identical copy of the data object published by the originating WIS2 Node. The simplest strategy is to accept the first notification message and download the data from the Global Cache instance that the message refers to by using a URL for the data object at that Global Cache instance. Alternatively, data consumers may have a preferred Global Cache instance, for example, one that is located in their region. Whichever Global Cache instance is chosen, data consumers will need to implement logic to discard duplicate notification messages based on `id` and duplicate data objects based on `data_id`.

A notification message also provides a small amount of metadata about the data object it references, such as location or time. Data consumers can use these metadata to decide whether the data object referenced in the message should be downloaded. This is known as client-side filtering.

The notification message should also include the metadata identifier for the dataset to which the data object belongs. A data consumer can use the metadata identifier to search the Global Discovery Catalogue and discover more about the data - in particular, whether there are any conditions on the use of those data.

1.2.4 How to download data

Links to where data can be accessed are made available through dataset discovery metadata (via the Global Discovery Catalogue) and/or data notification messages (via Global Brokers). Links can be used to directly download the data (according to the network protocol and content description provided in the link) using a mechanism appropriate to the workflow of the data consumer. Such mechanisms could include web and/or desktop applications, custom tools and so forth.

A discovery metadata record or notification message may provide more than one download link. The preferred link will be identified as "canonical" (link relation: "rel": "canonical" [5]).

Where data are provided through an interactive web service, a canonical link containing a URL from which data consumers can directly download a data object may be complemented with an additional link providing the URL for the root of the web service from which data consumers can interact with or query the entire dataset.

If a download link implements access control (for example, the data consumer needs to take some additional action(s) to download the data object), it will contain a security object that provides the pertinent information (such as the access control mechanism used and where/how a data consumer would need to register to request access).

1.2.5 How to use data

Data are shared on WIS2 in accordance with the WMO Unified Data Policy (Resolution 1 (Cg-Ext (2021))). This data policy describes two categories of data: core and recommended.

- Core data are considered essential for the provision of services for the protection of life and property and the well-being of all nations. Core data are provided on a free and unrestricted basis, without charge and with no conditions on use.
- Recommended data are exchanged on WIS2 in support of Earth system monitoring and prediction efforts. Recommended data may be provided with conditions on use and/or subject to a license.

The WMO Unified Data Policy (Resolution 1 (Cg-Ext (2021))) encourages attribution of the source of the data in all cases. This ensures that, credit is given to those who have expended effort and resources in collecting, curating, generating, or processing the data. Attribution provides visibility into who is using the data, which, for many organizations, serves as crucial evidence to justify the continued provision and updating of the data.

Details of the applicable WMO data policy and any rights or licenses associated with the data are provided in the discovery metadata accompanying the data. Discovery metadata records are available from the Global Discovery Catalogue.

The *Manual on WIS*, Volume II – Appendix F. WMO Core Metadata Profile (Version 2), 1.18 Properties / WMO Data Policy provides details on how the WMO Data Policy, rights and/or licenses are described in the discovery metadata.

When using data from WIS2, data consumers:

- Shall respect the conditions of use applicable to the data as expressed in the WMO Data Policy, rights statements, or licenses;
- Should attribute the source of the data.

1.2.6 Further reading for data consumers

Data consumers wanting to use data published via WIS2 should, at a minimum, read the following sections:

- PART I. Introduction
- 2.1 WIS2 architecture
- 2.2 Roles in WIS2
- 2.4 WIS2 Components

The following specifications in the *Manual on WIS*, Volume II also provide useful information:

- Appendix D. WIS2 Topic Hierarchy;
- Appendix E. WIS2 Notification Message;
- Appendix F. WMO Core Metadata Profile (Version 2).

1.3 Information for the data publisher

Data publishers wanting to share authoritative Earth system data with the WMO community should read the guidance presented here. A list of references to informative material in this Guide and elsewhere is provided at the end of this section.

1.3.1 How to get started

The first step is to consider the data, how they can be conceptually grouped into one or more datasets (see 1.1.4 Why are datasets so important?), and whether they are core or recommended data, as per the WMO Unified Data Policy (Resolution 1 (Cg-Ext(2021))) .

Next, it is important to consider where the data are published. If the data relate to a specific country or territory, they should be published through a National Centre (NC). If they relate to a region, programme, or other specialized function within WMO, they should be published through a Data Collection or Production Centre (DCPC). The functional requirements for NCs and DCPCs are described in the *Manual on WIS*, Volume II - Part III Functions of WIS.

All NCs and DCPCs are affiliated with a Global Information System Centre (GISC), which is responsible for helping to establish efficient and effective data sharing on WIS. The affiliated GISC can assist in getting the data onto WIS2.

It may be possible to identify an existing NC or DCPC that can publish the data. Alternatively, it may be necessary to establish a new NC or DCPC. The main distinction between these two centres is that an NC is designated by a Member, whereas a DCPC is designated by a WMO or related international programme and/or a regional association.

Both NCs and DCPCs require the operation of a WIS2 Node (see 2.4.1 WIS2 Node). The procedure for registering a new WIS2 Node is provided in 2.6.1.1 Registration and decommissioning of a WIS2 Node.

Once the scope of the datasets has been determined, the applicable data policy has been identified, and a WIS2 Node is ready for data publication, the process can proceed to the next step: providing discovery metadata.

1.3.2 How to provide discovery metadata to WIS2

Discovery metadata is the mechanism by which data publishers tell potential consumers about their data, how it may be accessed, and any conditions they may place on the use of those data.

Each dataset that is published must have an associated discovery metadata record. This record is encoded as GeoJSON (See RFC 7946[6]) and must conform to the specification given in the *Manual on WIS*, Volume II - Appendix F. WMO Core Metadata Profile (Version 2).

Copies of all discovery metadata records from WIS2 are held in the Global Discovery Catalogues, where data consumers can search and browse to find data that is of interest to them.

Depending on local arrangements, your GISC may be able to assist in transferring discovery metadata record(s) to the Global Discovery Catalogues. If this is not the case, data publishers will need to publish the discovery metadata record(s) themselves[7] using one of two methods:

- The simplest method is to encode the discovery metadata record as a file and publish it to an HTTP server, where it can be accessed with a URL.
- Alternatively, a data publisher may operate a local metadata catalogue through which discovery metadata records can be shared using an API (for example, OGC API – Records). Each discovery metadata record (for instance, an item that is part of the discovery metadata catalogue) can be accessed with a unique URL via the API.

In both cases, a notification message needs to be published on a Message Broker that tells WIS2 that there is a new discovery metadata record to upload and that it can be accessed at the specified URL.[8] Notification messages shall conform to the specification given in the *Manual on WIS*, Volume II - Appendix E. WIS2 Notification Message. They must also be published on a topic that conforms to the specification given in the *Manual on WIS*, Volume II - Appendix D. WIS2 Topic Hierarchy. For example, metadata published by Deutscher Wetterdienst would use the following topic: `origin/a/wis2/de-dwd/metadata`.

These discovery metadata records are then propagated through the Global Service components into the Global Discovery Catalogue, where data consumers can search and browse for datasets of interest.

Upon receipt of a new discovery metadata record, a Global Discovery Catalogue (see 2.4.4 Global Discovery Catalogue) will validate, assess, ingest, and publish the record. Validation ensures compliance with the specification, while the assessment evaluates the discovery record against good practices. The Global Discovery Catalogue will notify the data publisher if the discovery metadata record fails validation and provide recommendations for improvements.

Discovery metadata must be published in the Global Discovery Catalogues before the data are published.

Discovery metadata should be re-published on a daily basis.

1.3.3 How to provide data to WIS2

WIS2 is based on the web architecture.[9] As such it is *resource oriented*. Datasets are resources; the "granules" of data grouped in a dataset are resources; and the discovery metadata records that describe datasets are resources. In web architecture, every resource has a unique identifier (such as a URI[10]), which can be used to resolve the identified resource and interact with it (for example, to download a representation of the resource over an open-standard protocol such as HTTP).

In simple terms, data (and metadata) are provided to WIS2 by assigning them a unique identifier, in this case a URL[11], and making them available via a data server - most typically a web server using HTTP protocol.[12] It is up to the data server to decide what to provide when resolving the identifier. For example, the URL of a data granule may resolve as a representation encoded in a given data format, whereas the URL of a dataset may resolve as a description of the dataset (that is, metadata) that includes links to access the data from which the dataset is comprised - either individual files (that is, the data granules) or an interactive API that enables users to request only the parts of the dataset they need by specifying query parameters.

The following sections cover specific considerations relating to publishing data to WIS2.

1.3.3.1 Data formats and encodings

Whether providing data as files or through interactive APIs, data publishers need to decide which encodings (data formats) to use. WMO technical regulations may require that data be encoded in specific formats. For example, synoptic observations must be encoded in Binary Universal Form for the Representation of meteorological data (BUFR). The *Manual on Codes* (WMO-No. 306) provides details of data formats formally approved for use in WMO. However, the technical regulations do not cover all data sharing requirements. In such cases, data publishers should select data formats that are open, non-proprietary, widely adopted, and understood in the target user community. In this context, “open” means that anyone can use the format without needing a license – either to encode data in that format or to write software that understands it.

1.3.3.2 Providing data as files

The simplest way to publish data through WIS2 is to persist the data as files and publish those files on a web server. All these files need to be organized in some manner, for example, in a flat structure or grouped into collections that resemble folders or directory structures.

To ensure that the data are usable, users need to be able to find the specific file (or files) they need.

Naming conventions for files and/or directories are useful – but only if they are understood. If users do not understand the naming convention, it will be a barrier to widespread reuse, as many users will simply treat the filename as an opaque string. Where file naming conventions (such as names with embedded metadata) are commonly used by communities, they should only be used when adequate documentation is provided to users.

WIS2 does not require the use of specific naming conventions.

Another approach to enhance the usability of the data is to complement the collections (such as directories or folders in which files are grouped) with information that describes their content. Then users, both humans and software agents, can browse the structure and find what they need. Examples of this approach include:

- Web Accessible Folders (WAF) and README files: A web-based folder structure listing the data object files by name, where each folder contains a formatted README file describing the folder contents;

- SpatioTemporal Asset Catalog (STAC)[13]: A community standard based on GeoJSON to describe geospatial data files that can be easily indexed, browsed and accessed. Free and open source tools present STAC records (one for each data object file) through a web-based, browsable user interface.

When publishing collections of data, it is tempting to package content into zip or submission information package (SIP)[14] resources - perhaps even to package the entire collection, including folders, into a single resource. Similarly, WMO formats such as GRIB and BUFR allow multiple data objects (such as fields or observations) to be packed into a single file. Downloading a single resource is convenient for many users, but the downside is that the user must download the entire resource and then unpack/decompress it. The convenience of downloading fewer resources must be balanced against the cost of forcing users to download data they may not need. The decision should be guided by common practice in the specific domain - for example, only using zip files, SIP resources, or packing files if this is what the users expect.

1.3.3.3 Providing interactive access to data with APIs

Interactive data access aims to support efficient data workflows by enabling client applications to request only the data they need. The advantage of interactive data access is that it provides greater flexibility. Data publishers can offer an API structured around how users want to work with the data rather than forcing them to work with the structure that is convenient for the data publisher.

However, interactive data access is complex to implement. It requires a server running software that can:

1. Interpret a user’s request;
2. Extract the data from wherever they are stored;
3. Package those data and send them back to the user.

Importantly, when considering the use of interactive APIs to serve data, it is necessary to plan for costs: every request to an interactive API requires computational resources to process.

Based on the experience of data publishers that have been using web APIs to serve their communities, this Guide makes the following recommendations regarding interactive APIs:

- First, interactive APIs should be self-describing. Data consumers should not need to know, a priori, how to make requests from an API. They should be able to discover this information from the API endpoint itself – even if this simply entails a link to a documentation page they need to read.
- Second, APIs should comply with OpenAPI[15] version 3 or later. OpenAPI provides a standardized mechanism to describe the API. Tooling (free, commercial, etc.) that can read this metadata and automatically generate client applications to query the API is widely available.
- Third, the OGC has developed a suite of APIs[16] (called "OGC APIs") that are specifically designed to provide APIs for geospatial data workflows (discovery, visualization, access, processing/exploitation) – all of which build on OpenAPI. Among these, OGC API – Environmental Data Retrieval (EDR)[17], OGC API – Features[18], and OGC API - Coverages[19] are

considered particularly useful. Because these are open standards, there is an ever-growing suite of software implementations (both free and proprietary) that support them. It is recommended that data publishers assess these open-standard API specifications to determine their suitability for publishing their datasets using APIs.

Finally, it is advisable to consider versioning the API to avoid breaking changes when adding new features. A common approach is to add a *version number* prefix into the API path, for example, `/v1/service/{rest-of-path}` or `/service/v1/{rest-of-path}`.

More guidance on the use of interactive APIs in WIS2 is anticipated in future versions of this Guide.

1.3.3.4 Providing data in (near) real time

WIS2 is designed to support the data sharing needs of all WMO disciplines and domains. Among these, the World Weather Watch [20] drives specific needs for the rapid exchange of data to support weather forecasting.

To enable real-time data sharing[21] WIS2 uses notification messages to inform users of the availability of a new resource, either data or discovery metadata, and how they can access that resource. Notification messages are published to a queue on a Message Broker in a data publisher's WIS2 Node[22] using the MQTT protocol and immediately delivered to all users subscribing to that queue. A queue is associated with a specific *topic*, such as a dataset.

For example, when a new temperature profile from a radiosonde deployment is added to a dataset of upper-air data measurements, a notification message will be published that includes the URL used to access the new temperature profile data. All subscribers to notification messages about the upper-air measurement dataset will receive the notification message and be able to identify the URL and download the new temperature profile data.

Optionally, data may be embedded in a notification message using a `content` object in addition to being published via the data server. Inline data must be encoded as UTF-8, Base64, or gzip, and must not exceed 4096 bytes in length once encoded.

Notification messages are encoded as GeoJSON (RFC 7946) and must conform to the *Manual on WIS*, Volume II, Appendix E. WIS2 Notification Message.

The URL used in the notification message should refer only to the newly added data object (for example, the new temperature profile), rather than the entire dataset. However, the WIS2 Notification Message specification allows for multiple URLs to be provided. When providing data through an interactive API, it may be useful to provide a "canonical" link (designated by link relation: `"rel": "canonical"`[23]) and an additional link with the URL for the root of the web service from which the entire dataset can be accessed or queried.

The dataset identifier should be included in the notification message (`metadata_id` property). This allows data consumers receiving the notification to cross reference it with information provided in the discovery metadata for the dataset, for example the conditions of use specified in the data policy, rights, or license.

If controlled access to the data has been implemented (for example, the use of an API key), the download link should include a security object which provides the pertinent information (the access control mechanism used, where or how a data consumer needs to register to request access, and so forth).

To ensure that data consumers can easily find the topics they want to subscribe to, data publishers must publish to an authorized topic, as specified in the *Manual on WIS*, Volume II, Appendix D. WIS2 Topic Hierarchy.

If the data seem to relate to more than one topic, the most appropriate one should be selected. The topic hierarchy is not a knowledge organization system – it is used solely to ensure the uniqueness of topics for publishing notification messages. Discovery metadata is used to describe a dataset and its relevance to additional disciplines; each dataset is mapped to one, and only one, topic.

If the WIS2 Topic Hierarchy does not include a topic appropriate for the data, the data should be published on an experimental topic. This will allow data exchange to be established while the formalities are being considered.[24] Experimental topics are provided for each Earth system discipline at level eight in the topic hierarchy (for example, `origin/a/wis2/{centre-id}/data/{earth-system-discipline}/experimental/`). Data publishers can extend the experimental branch with subtopics they deem appropriate. Experimental topics are subject to change and will be removed once they are no longer needed. For more information, see *Manual on WIS*, Volume II, Appendix D. WIS2 Topic Hierarchy, section 1.2 Publishing.

Whatever topic is used, the discovery metadata provided to the Global Discovery Catalogue must include subscription links using that topic.[25] The Global Broker will only republish notification messages on topics specified in the discovery metadata records.

1.3.3.5 Considerations when providing core data in WIS2

Core data, as specified in the WMO Unified Data Policy (Resolution 1 (Cg-Ext(2021))) are considered essential for the provision of services for the protection of life and property and for the well-being of all nations. Core data is provided on a free and unrestricted basis, without charge and with no conditions on use.

WIS2 ensures highly available, rapid access to *most* core data via a collection of Global Caches (see 2.4.3 Global Cache). Global Caches subscribe to notification messages about the availability of new core data published at WIS2 Nodes, download a copy of that data and republish it on a high-performance data server and then discard it after the retention period expires (normally after 24 hours[26]). Global Caches do not provide sophisticated APIs. They publish notification messages advertising the availability of data on their caches and allow users to download data via HTTPS using the URL in the notification message.

The URL included in a notification message that is used to access core data from a WIS2 Node, or the "canonical" URL, if multiple URLs are provided, must:

1. Refer to an individual data object; and
2. Be directly resolvable, such that the data object can be downloaded simply by resolving the given URL without further action.

A Global Cache will download and cache the data object accessed via this URL.

The Global Caches are designed to help Members efficiently share real-time and near-real-time data. They ensure that core data are available to all on a free and unrestricted basis, as required by the WMO Unified Data Policy (Resolution 1 (Cg-Ext(2021))).

Unfortunately, Global Caches cannot republish *all* core data; there is a limit to how much data they can afford to serve. Currently, a Global Cache is expected to cache about 100 GB of core data each day.

If frequent updates to a dataset are very large (for example, in the case of weather prediction models or remote sensing observations) data publishers will need to share the burden of distributing their data with Global Cache operators. They should work with their GISC to determine the highest priority elements of their datasets that will be republished by the Global Caches.

Core data that are not to be cached must have the cache property in the notification message set to false.[27]

Data publishers must ensure that core data that are not cached are publicly accessible from their WIS2 Node, that is, with no access control mechanisms in place.

Global Cache operators may choose to disregard a cache preference, for example, if they feel that the content being providing is large enough to impede the provision of caching services for other Members.[28] In such cases, the Global Cache operator will log this behaviour. Global Cache operators will collaborate with data publishers and their GISCs to resolve any concerns.

Finally, note that Global Caches are under no obligation to cache data published on *experimental* topics. For such data, the `cache` property should be set to `false`.

1.3.3.6 Implementing access control

Recommended data, as defined in the WMO Unified Data Policy (Resolution 1 (Cg-Ext(2021))), are exchanged on WIS2 in support of Earth system monitoring and prediction efforts and may be provided with conditions on use. This means that the data publisher may control access to recommended data.

Access control should only use the "security schemes" for authentication and authorization specified in OpenAPI.[29]

Where access control is implemented, a `security` object should be included in the download links in discovery metadata and notification messages to provide the user with pertinent information about the access control mechanism used and where/how they might register to request access.

Recommended data are never cached by the Global Caches.

The use of core data must always be free and unrestricted. However, it may be necessary to leverage existing systems with built-in access control when implementing the download service for the WIS2 Node.

Example 1: API key. The data server requires a valid API key to be included in download requests. The URLs used in notification messages should include a valid API key.[30][31]

Example 2: Presigned URLs. The data server uses a cloud-based object store that requires credentials to be provided when downloading data. The URLs used in notification messages should be *presigned* with the data publisher's credentials and valid for the cache retention period (for example, 24 hours).[32]

In both cases, the URL provided in a notification message can be directly resolved without requiring a user or a Global Cache to take additional action, such as providing credentials or authenticating.

Finally, note that if only core data are being published, it may be possible to rely entirely on the Global Caches to distribute the data. In such cases, the WIS2 Node may use Internet Protocol (IP) filtering to allow access only from Global Services. For more details, see 2.6 Implementation and operation of a WIS2 Node.

1.3.3.7 Providing access to data archives

There is no requirement for a WIS2 Node to publish notification messages about newly available data; however, the mechanism is available if needed (for instance, for real-time data exchange). Data archives published via WIS2 do not need to provide notification messages for data unless the user community has expressed a need to be rapidly notified about changes (for example, the addition of new records to a climate observation archive).

However, notification messages must still be used to share discovery metadata with WIS2. Given that the provision of metadata and subsequent updates are likely to be infrequent, it may be sufficient to manually author notification messages as needed and publish them locally on an MQTT broker[33] or with the help of a GISC. See above for more details on publishing discovery metadata to WIS2.

Note that some data archives, for example, Essential Climate Variables, are categorized as core data. Core data may be distributed via the Global Caches; however, given that they provide only short-term data hosting (for instance, for 24 hours), Global Caches are not an appropriate mechanism to provide access to core data archives. These archives must be accessed directly via the WIS2 Node.

1.3.4 Further reading for data publishers

Data publishers planning to operate WIS2 Nodes, at a minimum, should read the following sections:

- PART I. Introduction
- 2.1 WIS2 architecture
- 2.2 Roles in WIS2
- 2.4 WIS2 Components
- 2.6 Implementation and operation of a WIS2 Node

The following sections are recommended for further reading:

- [_part_iii_information_management]
- [_part_iv_security]
- [_part_v_competencies]

Note that *PART V. Security* and *PART VI. Competencies* reference content originally published for

WIS1. These sections remain largely applicable and will be updated in subsequent releases of this Guide.

Data publishers publishing aviation weather data via WIS2 for onward transmission through the International Civil Aviation Organization (ICAO) System Wide Information Management (SWIM), should also read 2.8.1.1 Publishing meteorological data through WIS2 into ICAO SWIM.

Finally, data publishers should also review the specifications in the *Manual on WIS*, Volume II:

- Appendix D. WIS2 Topic Hierarchy
- Appendix E. WIS2 Notification Message
- Appendix F. WMO Core Metadata Profile (Version 2)

[1] See *Data Catalog Vocabulary (DCAT) – Version 3, W3C Recommendation 22 August 2024* https://www.w3.org/TR/vocab-dcat-3/#Class:Dataset

[2] In this example, the system used to publish the data only retains the data for five days. Other systems may retain the data for a longer or shorter period of time.

[3] Subscribing to notifications about newly available data ensures that the data consumers do not need to continually to poll the data server to check for updates.

[4] Internet search engines allow data consumers to discover WIS2 datasets by indexing the content in Global Discovery Catalogues.

[5] See Internet Assigned Numbers Authority (IANA) Link Relations: https://www.iana.org/assignments/link-relations/link-relations.xhtml

[6] See RFC 7946 - The GeoJSON Format: https://datatracker.ietf.org/doc/html/rfc7946.

[7] In the future, WIS2 may provide metadata publication services (for example, through a WIS2 metadata management portal) to assist with this task. However, such services are not currently available.

[8] Both data and metadata are published using the same notification message mechanism to announce the availability of new resources.

[9] See Architecture of the World Wide Web, Volume One: https://www.w3.org/TR/webarch/.

[10] See RFC 3986 - Uniform Resource Identifier (URI) - Generic Syntax: https://datatracker.ietf.org/doc/html/rfc3986.

[11] The term "Uniform Resource Locator" (URL) refers to the subset of URIs that, in addition to identifying a resource, provide a means of locating the resource by describing its primary access mechanism (such as its network location). See RFC 3986: https://datatracker.ietf.org/doc/html/rfc3986.

[12] WIS2 strongly prefers secure versions of protocols (such as HTTPS), wherein the communication protocol is encrypted using Transport Layer Security (TLS).

[13] See STAC: SpatioTemporal Asset Catalogs: https://stacspec.org/en.

[14] See https://www.iasa-web.org/tc04/submission-information-package-sip or https://user.eumetsat.int/resources/user-guides/formats.

[15] See OpenAPI Specification v3.1.0: https://spec.openapis.org/oas/v3.1.0.

[16] See OGC API: https://ogcapi.ogc.org/.

[17] See OGC API - Environmental Data Retrieval (EDR): https://ogcapi.ogc.org/edr.

[18] See OGC API - Features: https://ogcapi.ogc.org/features.

[19] See OGC API - Coverages: https://ogcapi.ogc.org/coverages.

[20] See World Weather Watch: https://wmo.int/world-weather-watch.

[21] In the context of WIS2, real time implies anything from a few seconds to a few minutes - not the milliseconds required by some applications.

[22] WIS2 ensures the rapid global distribution of notification messages using a network of Global Brokers which subscribe to the Message Brokers of WIS2 Nodes and republish notification messages (see [_2_4_2_Global_Broker]).

[23] See Internet Assigned Numbers Authority (IANA) Link Relations: https://www.iana.org/assignments/link-relations/link-relations.xhtml.

[24] Experimental topics are necessary for the WIS2 pre-operational phase and future pre-operational data exchange in test mode.

[25] The Global Discovery Catalogue will reject discovery metadata records containing links to topics outside the official topic hierarchy.

[26] A Global Cache provides short-term hosting of data. Consequently, it is not an appropriate mechanism to provide access to archives of core data, such as Essential Climate Variables. Providers of such archive data must be prepared to serve such data directly from their WIS2 Node.

[27] The default value for the `cache` property is `true`. Omitting the property will result in the data object being cached.

[28] Excessive data volume is not the only reason a Global Cache operator may refuse to cache content. Other reasons include too many small files, unreliable download from a WIS2 Node, and so forth.

[29] See OpenAPI Security Scheme Object: https://spec.openapis.org/oas/v3.1.0#security-scheme-object.

[30] A specific API key should be used for the publication of data via WIS2 so that data usage can be tracked.

[31] Given that users are encouraged to download core data from the Global Cache, there will likely be limited access using the API key of the WIS2 account. If the usage quota for the WIS2 account is exceeded (for instance, if further data access is blocked), users should download via the Global Cache as mandated in the *Manual on WIS*, Volume II.

[32] See working with presigned URLs on Amazon S3: https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html.

[33] MQTT broker managed services are available online, often with a free starter plan sufficient for the occassional publication of notifications about metadata. These services provide a viable alternative to implementing an MQTT broker instance.

PART II. Architecture, implementation and operations

2.1 WIS2 architecture

WIS2 is a federated system of systems based on web architecture and open standards, comprising many WIS2 Nodes, for publishing data and Global Services that enable fault tolerant, highly available, low-latency data distribution.

NCs, DCPCs, and GISCs are all types of WIS centres.

NCs and DCPCs operate WIS2 Nodes.

GISCs coordinate the operation of WIS within their area of responsibility (AoR) and ensure the smooth operation of the WIS2 system.

A WIS centre may also operate one or more Global Services.

WIS centres shall comply with the technical regulations defined in the *Manual on WIS*, Volume II.

2.2 Roles in WIS2

When describing the functions of WIS2, there are four roles to consider:

1. Data publisher;
2. Global coordinator;
3. Global Service operator;
4. Data consumer.

These roles are outlined below.

2.2.1 Data publisher

- This role is fulfilled by NCs and DCPCs.
- Data publishers operate a WIS2 Node to share their data within the WIS2 ecosystem.
- Data publishers manage, curate, and provide access to one or more datasets.
- For each dataset, a data publisher provides:
    1. Discovery metadata to describe the dataset and provide details on how it can be accessed and under what conditions;
    2. An API or web service to access or interact with the dataset;
    3. Notification messages advertising the availability of new data and metadata.

2.2.2 Global coordinator

- This role is exclusive to GISCs.
- All GISCs supporting WMO Members in their AoR fulfil their data sharing obligations via WIS2.

2.2.3 Global Service operator

- To ensure highly available global data exchange, a WIS centre may operate one or more Global Services:
  1. The Global Discovery Catalogue enables users to search all datasets provided by data publishers and discover where and how to interact with those datasets (for example, to subscribe to updates, to access/download/visualize data, to access more detailed information about the dataset);
  2. A Global Broker provides highly available messaging services through which users may subscribe to notifications about all datasets provided by data publishers;
  3. A Global Cache provides a highly available download service for cached copies of core data downloaded from data publishers' web services;
  4. A Global Monitor gathers and displays system performance, data availability, and other metrics from all WIS2 Nodes and Global Services;

2.2.4 Data consumer

- This role represents anyone wanting to find, access, and use data from WIS2. Examples include NMHSs, government agencies, research institutions, private sector organizations, and so forth.
- Data consumers search or browse a Global Discovery Catalogue (or another search engine) to discover the datasets that meet their needs ("datasets of interest").
- Data consumers subscribe via a Global Broker to receive notification messages about the availability of data or metadata associated with their datasets of interest.
- Data consumers determine whether the data or metadata referenced in the notification messages are required.
- Data consumers download data from a Global Cache or WIS2 Node.

2.3 WIS2 specifications

Leveraging existing open standards, WIS2 defines the following specifications in support of publication, subscription, notification and discovery:

Table 1. WIS2 specifications

Specification	Granularity	Primary WIS2 Component(s)
WMO Core Metadata Profile 2 (WCMP2)	Datasets	Global Discovery Catalogue
WIS2 Topic Hierarchy (WTH)	Dataset granules	Global Broker, WIS2 Nodes

Specification	Granularity	Primary WIS2 Component(s)
WIS2 Notification Message (WNM)	Dataset metadata, dataset granules	Global Broker, WIS2 Nodes

Refer to the *Manual on WIS*, Volume II for details.

2.4 WIS2 Components

2.4.1 WIS2 Node

- WIS2 Nodes are central to WIS2. They are operated by NCs and DCPCs to publish their core and recommended data.
- WIS2 adopts web technologies and open standards, enabling WIS2 Nodes to be implemented using freely available software components and common industry practices.
- WIS2 Nodes publish data as files on a web server or using an interactive web service.
- WIS2 Nodes describe the data they publish using discovery metadata. See the *Manual on WIS*, Volume II, Appendix F. WMO Core Metadata Profile (Version 2).
- WIS2 Nodes generate notification messages advertising the availability of new data. These notification messages are published to a Message Broker. The WIS2 Topic Hierarchy is used to ensure that all WIS2 Nodes publish to consistent topics. The information in the notification message tells the data consumer the location from which to download the data. Notification messages are also used to inform data consumers of the availability of discovery metadata. See the Manual on WIS, Volume II, Appendix D. WIS2 Topic Hierarchy and Appendix E. WIS2 Notification Message.
- WIS2 Nodes may control access to the data they publish. Global Services operate with fixed IP addresses, enabling WIS2 Nodes to easily distinguish their requests.

2.4.2 Global Broker

- WIS2 incorporates several Global Brokers, ensuring the highly resilient distribution of notification messages across the globe.
- A Global Broker subscribes to the Message Broker operated by each WIS2 Node and republishes notification messages.
- A Global Broker subscribes to notifications from other Global Brokers to ensure that it receives a copy of all notification messages.
- A Global Broker republishes notification messages from every WIS2 Node and Global Service.
- A Global Broker operates a highly available, high-performance Message Broker.
- A Global Broker uses the WIS2 Topic Hierarchy, enabling data consumers to easily find topics relevant to their needs.
- Data consumers should subscribe to notifications from a Global Broker, not directly from the Message Brokers operated by WIS2 Nodes.

2.4.3 Global Cache

- WIS2 incorporates several Global Caches, ensuring the highly resilient distribution of data across the globe.
- A Global Cache provides a highly available data server, from which a data consumer can download core data, as specified in Resolution 1 (Cg-Ext(2021)).
- A Global Cache subscribes to notification messages via a Global Broker.
- Upon receiving a notification message, the Global Cache downloads a copy of the data referenced in the message from the WIS2 Node, makes these data available on its server, and publishes a new notification message informing data consumers that they can now access these data on its server.
- A Global Cache will subscribe to notification messages from other Global Caches, enabling it to download and republish data that it has not acquired directly from WIS2 Nodes. This ensures that each Global Cache holds data from every WIS2 Node.
- A Global Cache shall retain a copy of the core data for a duration compatible with the real-time or near-real-time schedule of the data and not less than 24 hours.
- A Global Cache will delete data from the cache once the retention period has expired.
- Data consumers should download data from a Global Cache when those data are available.

2.4.4 Global Discovery Catalogue

- WIS2 includes several Global Discovery Catalogues.
- A Global Discovery Catalogue enables a data consumer to search and browse descriptions of data published by each WIS2 Node. The data description (discovery metadata) provides sufficient information to determine the usefulness of the data and how it may be accessed.
- A Global Discovery Catalogue subscribes to notification messages about the availability of new (or updated) discovery metadata via a Global Broker. It downloads a copy of the discovery metadata and updates the catalogue.
- A Global Discovery Catalogue amends discovery metadata records to add details of where one can subscribe to updates about the dataset at a Global Broker.
- A Global Discovery Catalogue makes its content available for indexing by search engines.

2.4.5 Global Monitor

- WIS2 includes a Global Monitor service.
- The Global Monitor collects metrics from WIS2 components.
- The Global Monitor provides a dashboard that supports the operational management of the WIS2 system.
- The Global Monitor tracks:
    1. What data is published by WIS2 Nodes;
    2. Whether the data can be effectively accessed by data consumers;
    3. The performance of components in the WIS2 system.

2.5 Protocol configuration

2.5.1 Publish-subscribe protocol (MQTT)

- The MQTT protocol[1] is to be used for all WIS2 publish-subscribe workflows (publication and subscription).
- MQTT v3.1.1 and v5.0 are the chosen protocols for the publication of and subscription to WIS2 notification messages.
    - MQTT v5.0 is preferred for connecting to Global Brokers as it provides additional features such as the ability to use shared subscriptions.
- The following parameters are to be used for all MQTT client/server connections and subscriptions:
    - Message retention: false;
    - Quality of Service (QoS) of 1;
    - A maximum of 2000 messages to be held in a queue per client.
- To enable user authentication and authorization, WIS2 Nodes, Global Caches, Global Discovery Catalogues and Global Brokers shall use a user and password based mechanism.
- To improve the overall level of security of WIS2, the secure version of the MQTT protocol is preferred. If used, the certificate must be valid.
- The standard Transmission Control Protocol (TCP) ports to be used are 8883 for Secure MQTT (MQTTS) and 443 for Secure Web Socket (WSS).

2.5.2 Download protocol (HTTP)

- The HTTP protocol (RFC 7231[2]) is to be used for all WIS2 download workflows.
- To improve the overall level of security of WIS2, the secure version of the HTTP protocol is preferred. If used, the certificate must be valid.
- The standard TCP port to be used is 443 for Secure HTTP (HTTPS).

2.6 Implementation and operation of a WIS2 Node

2.6.1 Practices and procedures

2.6.1.1 Registration and decommissioning of a WIS2 Node

The registration and decommissioning of WIS2 Nodes must be approved by the Permanent Representative (PR) with WMO of the country or territory where the WIS centre is located. The WIS National Focal Point (NFP) can register a WIS2 Node on behalf of the PR for an official NC or DCPC listed in the Manual on WIS, Volume I. Where the WIS2 Node is part of a DCPC, the sponsoring WMO programme or regional association shall be consulted.

A WIS2 Node can be registered to exchange data concerning a WMO project or campaign for a limited time. The WIS NFP can register such a project-related WIS2 Node in coordination with the

WMO Secretariat.

A WIS2 Node can act as a publication facility on behalf of other centres. This is a Data Collection or Production Centre (DCPC) role, as defined in the Manual on WIS. Data or metadata publication by a DCPC will use the centre identifiers of the data producers.

The WMO Secretariat will maintain a WIS2 register with an authoritative list of WIS2 Nodes and Global Services.

The registration of a WIS2 Node involves the following steps:

- Request to host a WIS2 Node: A request to host a WIS2 Node shall be put forward by the WIS NFP of the country of the WIS2 Node host centre, or, in the case of international organizations, by either the PR with WMO of the country or territory where the centre is located or the president of the relevant organization, if the WMO partner or programme is designated as a DCPC.
- Assign a centre identifier ("centre-id"): The centre-id is an acronym proposed by the Member and endorsed by the WMO Secretariat. It is a single identifier consisting of a top-level domain (TLD) and a centre name and represents the data publisher, distributor or issuing centre of a given dataset or data product/granule (see the Manual on WIS, Volume II – Appendix D. WIS2 Topic Hierarchy). See (2.6.1.2 Guidance on assigning a centre identifier for a WIS2 Node).
- Complete the WIS2 register: The WIS NFP shall complete the WIS2 register maintained by the WMO Secretariat.
- Provide details of the Global Service: The WMO Secretariat provides connection details (such as IP addresses) for the Global Services so that the WIS2 Node can be configured to provide access.
- WIS2 Node assessment: The principal GISC verifies that the WIS2 Node is compliant with WIS2 requirements. This assessment includes:
    - Verification of compliance of the topics used by the centre with the WTH specification;
    - Verification of compliance of notification messages with the WNM specification;
    - Verification that the data server is correctly configured and properly functioning;
    - Verification that the Message Broker is correctly configured and properly functioning.
- Add a new centre to WIS2: Upon completion of the verification and confirmation that the centre satisfies all the conditions for operating a WIS2 Node, the GISC notifies the WMO Secretariat and confirms that the centre can be added to WIS2 as a WIS2 Node.
- Communicate the details to the Global Services: The WMO Secretariat provides the details of the WIS2 Node to the Global Brokers so that they can subscribe to the WIS2 Node.

A diagram of the process for registering a WIS2 Node is presented below (see Figure 1).

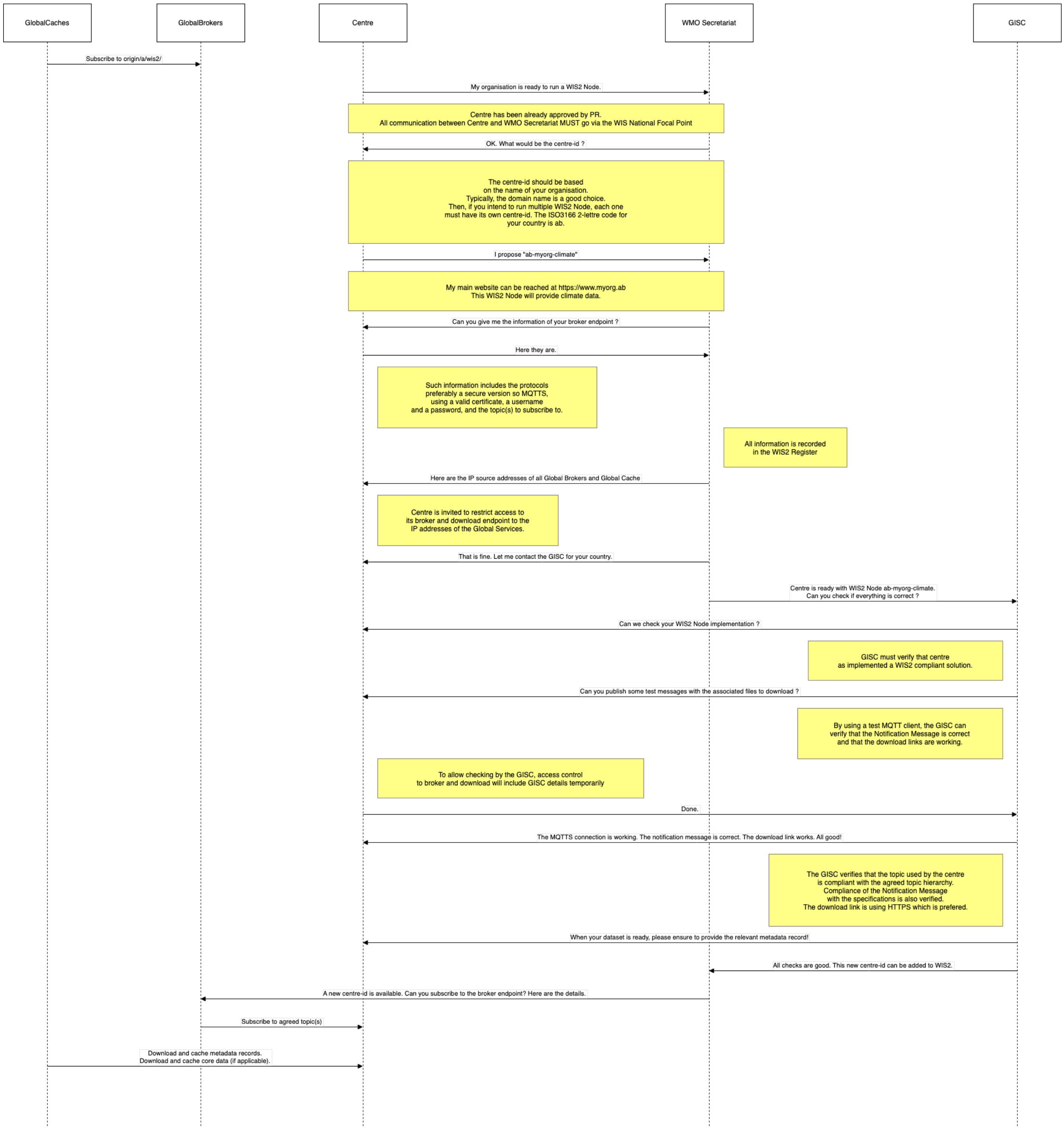

Figure 1. Diagram of the process for registering a WIS2 Node

Once a WIS2 Node has been registered and connected to the Global Services, it can proceed to register the datasets it will publish via WIS2. To register a dataset, the WIS2 Node publishes discovery metadata about the new dataset. Validation of the discovery metadata is completed by the Global Discovery Catalogues, and the Global Brokers automatically subscribe to the topics provided in the discovery metadata record. For more information, see 1.3.2 How to provide discovery metadata to WIS2.

Once the dataset has been successfully registered, the WIS2 Node can proceed to exchange data - see 1.3.3 How to provide data to WIS2.

When decommissioning a WIS2 Node, operators must ensure that obligations relating to data sharing within WIS continue to be met after the WIS2 Node is decommissioned, for example, by migrating the data sharing obligations to another WIS2 Node. In the case of a DCPC, this may mean transferring the responsibilities to another Member.

2.6.1.2 Guidance on assigning a centre identifier for a WIS2 Node

The centre identifier (`centre-id`) is used in WIS2 to uniquely identify a participating WIS2 Node. The `centre-id` must conform to the specifications given in the *Manual on WIS*, Volume II - Appendix D. WIS2 Topic Hierarchy, section 7.1.6 Centre identification.

The `centre-id` comprises two dash-separated tokens.

Token 1 is a Top Level Domain (TLD) defined by the Internet Assigned Numbers Authority (IANA).[3]

It is usually fairly easy for a Member to choose a TLD. However, for Members' overseas territories, this may require some thought. The recommended approach depends on the governance structure of the overseas territory. For example, Réunion is a French Department; it is considered part of France, and it uses the Euro. Réunion would use the "fr" TLD. New Caledonia is a French overseas territory with a TLD of "nc" because it has a separate, devolved governance structure. The recommendation is to use "nc". However, the decision of which TLD to use is made at the national level.

Token 2 is a descriptive name for the centre. It may contain dashes, but it may not contain other special characters.

The descriptive name should be something recognizable – not only by the WIS2 community, but also by other users. Basing the name on the web domain name is likely to ensure that centre identifiers remain unique within a particular country or territory. For example, the National Meteorological Service of the United Kingdom of Great Britian and Northern Ireland is the Met Office,[4] so "metoffice" is better than "ukmo".[5]. Using a four-letter GTS centre identifier (for example, CCCC) is not recommended because those who are unfamiliar with GTS will not understand these identifiers.

The centre identifier specification says that larger organizations operating multiple centres may wish to register separate centre-ids for each centre. This is a good practice. Keeping with the UK example, the Met Office operates a National Meteorological Centre (NMC), 9 DCPCs (for example, a Volcanic Ash Advisory Centre) and a WIS2 Global Service, so it is important to separate them. For example:

- `uk-metoffice-nmc`;
- `uk-metoffice-vaac`;
- `uk-metoffice-global-cache`.

It is not advisable to use a system name in the centre-id because system names may change over time. Functional designations are durable over the long term. Test WIS2 Nodes may be designated by adding "-test" to the descriptive name.

2.6.1.3 Authentication, authorization, and access control for a WIS2 Node

When configuring a WIS2 Node, it is necessary to consider how it will be accessed by Global Services and data consumers.

Global Brokers must authenticate when they connect to the MQTT Message Broker in the WIS2 Node. Username and password credentials are used.[6] When registering the WIS2 Node with the WMO Secretariat, these credentials must be provided. The WMO Secretariat will share the credentials with the Global Service operators and store them in the WIS register. These credentials should not be considered confidential or secret.

Given that Global Brokers republish notification messages provided by the WIS2 Node, access to the MQTT Message Broker may be restricted. Global Brokers operate using a fixed IP address, which allows access to be granted using IP filtering.[7] MQTT Message Brokers must be accessible by more than one Global Broker to ensure resilient transmission of notification messages to WIS2.

If your WIS2 Node only publishes core data[8], access to the data server may also be restricted, with the distribution of data handled by Global Caches. Global Caches also operate on fixed IP addresses, allowing their connections to be easily identified. Again, access must be granted to more than one Global Broker to ensure resilience.

During registration, the WMO Secretariat will provide host names and IP addresses of the Global Services to enable access controls to be configured.

Access controls may be implemented for recommended data. Only the security schemes for authentication and authorization specified in OpenAPI should be used.[9]

2.6.1.4 Improving Level of Service of a WIS Centre

It is possible to improve the level of service of a WIS centre by running a primary and a backup node. When the MQTT endpoint provided by the primary node is reachable, the Global Broker will subscribe to the primary node broker. When the MQTT connection is lost, the Global Broker will wait for one minute. If after one minute, the connection to the primary node broker is still down, the connection to the backup node will be activated. When the connection to the primary node is reestablished and stable for one minute, the connection to the backup node will be stopped.

It must be noted that the switch between the primary and the backup node is **only** controlled by the connectivity to the MQTT endpoint. The permanent absence of messages on the primary node or the failure to download core data from the Global Cache will not trigger the switch to the backup node.

It is suggested to run the two instances, primary and backup, as independently as possible (different hardware, different location – but same `centre-id` -) and permanently (the backup node should always publish the notification messages). This ensures the maximum possible level of service and limit the need for manual intervention to switch between primary and backup, or vice-versa.

2.6.2 Performance management

2.6.2.1 Service levels and performance indicators

A WIS2 Node must be able to publish datasets and compliant discovery metadata. This entails:

- Publishing metadata to the Global Data Catalogue;
- Publishing core data to the Global Cache;
- Publishing data for consumer access;
- Publishing data embedded in a message (for example, Common Alerting Protocol (CAP) warnings);
- Receiving metadata publication errors from the Global Data Catalogue;
- Providing metadata with topics to Global Brokers.

2.6.2.2 System performance metrics

If contacted by a Global Monitor for a performance issue via a GISC, the WIS2 Node should provide metrics to the GISC and the Global Monitor when service is restored to inform them of the resolution of the issue.

2.6.3 WIS2 Node Reference Implementation: WIS2 in a box

When providing a WIS2 Node, Members may use whichever software components they consider most appropriate to comply with the WIS2 technical regulations.

To assist Members, a free and open-source Reference Implementation called “WIS2 in a box” (wis2box) is available. wis2box implements the requirements for a WIS2 Node and contains additional enhancements. wis2box is built free and open-source software components that are mature, robust and widely adopted for operational use.

wis2box provides the functionality required for both data publisher and data consumer roles, as well as the following technical functions:

- Configuration, generation and publication of data (real-time or archive) and metadata to WIS2, compliant to WIS2 Node requirements
- MQTT Message Broker and notification message publication (subscribe);
- HTTP object storage and raw data access (download);
- Station metadata curation/editing tools (user interface);
- Discovery metadata curation/editing tools (user interface);
- Data entry tools (user interface);
- OGC API server, providing dynamic APIs for discovery, access, visualization and processing functionality (APIs);
- Extensible data "pipelines", allowing for the transformation, processing and publishing of additional data types;
- Provision of system performance and data availability metrics;
- Access control for publication of recommended data, as required;

- Subscription to notifications and download of WIS data from Global Services;
- Modular design, which allows for extensibility to meet additional requirements or integration with existing data management systems.

The project documentation can be found at https://docs.wis2box.wis.wmo.int.

wis2box is managed as a free and open source project. The source code, issue tracking and discussions are hosted openly on GitHub: https://github.com/wmo-im/wis2box.

2.7 Implementation and operation of a Global Service

2.7.1 Procedure for registering a new Global Service

The successful operation of WIS2 depends on a set of Global Services running well-managed IT environments with a very high level of reliability so that all WIS2 users and WIS2 Nodes are able to access and provide the data they need for their duties.

Depending on the nature of the Global Service, the following are the minimum capabilities needed to ensure that the level of service as a whole reaches 100% (or very close):

- Three Global Brokers, with each Global Broker connected to at least two other Global Brokers;
- Three Global Caches, with each Global Cache connected to at least two Global Brokers and capable of downloading data from all WIS2 Nodes providing core data;
- Two Global Discovery Catalogues, with each Global Discovery Catalogue connected to at least one Global Broker;
- Two Global Monitors - each Global Monitor should scrape the metrics from all other Global Services

In addition to the above, WIS architecture can accommodate adding (or removing) Global Services. Candidate WIS centres should inform their WIS NFP and contact the WMO Secretariat to discuss their offer to provide a Global Service.

Running a Global Service is a significant commitment for a WIS centre. To maintain a very high level of service, each Global Service has a key role to play.

On receipt of an offer from a Member to operate a Global Service, the WMO Secretariat will suggest which Global Service the Member may provide to improve WIS2. This suggestion will be based on the current situation of WIS2 (such as the number of existing Global Brokers, whether an additional Global Cache is needed, and so forth).

The *Manual on WIS*, Volume II, the present Guide, and other available materials will help WIS centres decide how to proceed.

When a decision on how to proceed has been made, the WIS NFP will inform the WMO Secretariat of its preference. Depending on the type of Global Service, the WMO Secretariat will provide a checklist to the WIS centre so that the future Global Service can be included in WIS operations.

A WIS centre must commit to running the Global Service for a minimum of four years.

The WMO Secretariat and other Global Services will make the required changes to include the new Global Service in WIS operations.

2.7.2 Performance management and monitoring of a Global Service

2.7.2.1 Monitoring and metrics for WIS2 operations

The availability of data and the performance of system components within WIS2 are actively monitored by GISCs and the Global Monitor service to ensure proactive responses to incidents and effective capacity planning for future operations.

WIS2 requires that metrics are provided using OpenMetrics[10] – the widely adopted, de-facto standard[11] for transmitting cloud-native metrics at scale. Many commercial and open-source software components already come preconfigured to provide performance metrics using the OpenMetrics standard. Tools such as Prometheus and Grafana aggregate and visualize metrics provided in this format, making it simple to generate performance insights.

WIS2 Global Services (Global Brokers, Global Caches, and Global Discovery Catalogues) provide monitoring metrics about their respective service to Global Monitors.

There is no requirement for WIS2 Nodes to provide monitoring metrics. However their WIS2 interfaces may be queried remotely by Global Services, which can then provide metrics on the availability of WIS2 Nodes.

Metrics for WIS2 monitoring should follow the naming convention `wmo_<program>_<name>`, where `<program>` is the name of the responsible WMO programme and `<name>` is the name of the metric. Examples of WIS2 metrics include (but are not limited to):

- `wmo_wis2_gc_downloaded_total`
- `wmo_wis2_gb_messages_invalid_total`

The full set of the WIS2 monitoring metrics is given in WMO: WIS2 Metric Hierarchy [12]

2.7.2.2 Service levels, performance indicators, and fair usage policies

- Each WIS centre operating a WIS2 Node is responsible for achieving the highest possible level of service based on its resources and capabilities.
- All Global Services, in particular Global Brokers and Global Caches, are collectively responsible for making WIS a reliable and efficient means of exchanging the data required for the operation of all WIS centres. The architecture provides a redundant solution where the failure of one component will not impact the overall level of service of WIS.
- Each Global Service should aim to achieve at least 99.5% availability of the service it provides. This is not a contractual target. It should be considered by the entity providing the Global Service as a guideline when designing and operating that service.
- Any Global Service operator anticipating an outage that may exceed 4 hours should notify other WIS2 centres (including other Global Services and the WMO Secretariat) in advance to allow them to prepare for the disruption.
- A Global Broker:

    - Should support a minimum of 200 WIS2 Nodes or Global Services;
    - Should support a minimum of 1 000 subscribers;
    - Should support the processing of a minimum of 10 000 messages per second.
- A Global Cache:
    - Should support a minimum of 100 GB of data in the cache;
    - Should support a minimum of 1 000 simultaneous downloads;
    - Could limit the number of simultaneous connections from a user (known by its originating source IP) to five;
    - Could limit the bandwidth usage of the service to 1 Gb/s.
- A Global Monitor:
    - Should support a minimum of 50 metrics providers;
    - Should support 200 simultaneous "access" to the dashboard;
    - Could limit the bandwidth usage of the service to 100 Mb/s.
- A Global Discovery Catalogue:
    - Should support a minimum of 20 000 metadata records;
    - Should support a minimum of 50 requests per second to the API endpoint.

2.7.2.3 Metrics for Global Services

In the following sections, and for each Global Service, a set of metrics is defined. Each Global Service will provide those metrics. They will then be ingested by the Global Monitor.

2.7.3 Global Broker

2.7.3.1 Technical considerations

- As detailed above, there will be at least three Global Brokers to ensure that messages within WIS2 are highly available and delivered globally with low latency.
- A Global Broker subscribes to messages from WIS2 Nodes and other Global Services. The Global Broker should aim to subscribe to all WIS centres. If this is not possible, the Global Broker should inform the WMO Secretariat so that the situation can be documented.
- Every WIS2 Node or Global Service must have subscriptions from at least two Global Brokers.
- For full global coverage, a Global Broker will subscribe to messages from at least two other Global Brokers.
- When subscribing to messages from WIS2 Nodes and other Global Services, a Global Broker must authenticate using the valid credentials managed by the WIS centre and available at WMO Secretariat.
- A Global Broker is built around two software components:
    - An off the shelf broker implementing both MQTT 3.1.1 and MQTT 5.0 in a highly available setup, typically in a cluster mode. Tools such as EMQX, HiveMQ, VerneMQ, RabbitMQ (in its latest versions) are compliant with these requirements. The open source version of

Mosquitto cannot be clustered and therefore should not be used as part of a Global Broker.

  - Additional features, including anti-loop detection, notification message format compliance, validation of the published topic, and metrics provision.

- When connected to the local WIS centre broker, the `wmo_wis2_gb_connected_flag` will be set to 1. When the connection to the local WIS centre broker is not possible, the `wmo_wis2_gb_connected_flag` will be set to 0.

- When receiving a message from a local WIS centre broker or a Global Service broker, the metric `wmo_wis2_gb_messages_received_total` will be increased by 1.

- A Global Broker will check if a discovery metadata record exists corresponding to the topic on which a message has been published. If there is no corresponding discovery metadata record, the Global Broker will discard non-compliant messages and will raise an alert. The metric `wmo_wis2_gb_messages_no_metadata_total` will be increased by 1. The Global Broker should not request information from a Global Discovery Catalogue for each notification message but should keep a cache of all valid topics for every `centre-id`.

- A Global Broker will check that the topic on which the message is received is valid. If the topic is invalid, the Global Broker will discard non-compliant messages and will raise an alert. The metric `wmo_wis2_gb_invalid_topic_total` will be increased by 1.

- Suggest to remove : During the pre-operational phase (2024), a Global Broker will not discard the message but instead will send a message on the `monitor` topic hierarchy to inform the originating centre and its GISC.~~

- A Global Broker will validate notification messages against the standard format (see *Manual on WIS*, Volume II – Appendix E. WIS2 Notification Message), discarding non-compliant messages and raising an alert. The metric `wmo_wis2_gb_invalid_format_total` will be increased by 1.

- A Global Broker will republish a message only once. It will record the message identifier (`id`) (as defined in the WIS2 Notification Message) of messages already published and will discard subsequent identical messages (those with the same message `id`). This is the anti-loop feature of the Global Broker.

- When publishing a message to the local broker, the metric `wmo_wis2_gb_messages_published_total` will be increased by 1.

- In order to improve the service level, a WIS centre may decide to operate two nodes sharing the same `centre-id`. One will be considered as *primary* and the other one as *backup*. In this case, when the primary WIS centre broker is reachable (the MQTT connection is up and running), the Global Broker will only process the notification messages from this primary node. When the connection is dropped (the MQTT connection is down) for at least one minute, the Global Broker will start processing the notification messages received from the backup WIS centre broker. When the connection to the primary WIS centre broker is active for at least one minute, the Global Broker will stop processing the messages coming from the backup WIS centre broker and will only process the messages received from the primary WIS centre broker. When the Global Broker is processing the messages from the backup WIS centre broker the metric `wmo_wis2_gb_connected_backup_flag` will be set to 1. When the backup WIS centre broker is defined but is not used as the source of notification message, the metric `wmo_wis2_gb_connected_backup_flag` will be set to 0.

- All above-defined metrics will be made available on HTTPS endpoints that the Global Monitor will ingest regularly.

- As a convention, the Global Broker centre-id will be `{tld}-{centre-name}-global-broker`.
- A Global Broker should operate with a fixed IP address so that WIS2 Nodes can permit access to download resources based on IP address filtering. A Global Broker should also operate with a publicly resolvable Domain Name System (DNS) name pointing to that IP address. The WMO Secretariat must be informed of the IP address and/or hostname and any subsequent changes.

2.7.4 Global Cache

In WIS2, Global Caches provide access to WMO core data for data consumers. This allows data providers to restrict access to their systems to Global Services, and it reduces the need for them to provide high bandwidth and low latency access to their data. Global Caches operate in a way that is transparent to end users in that they resend notification messages from data providers. These messages are updated to point to copies of the original data held in the Global Cache data store. Global Caches also resend notification messages from data providers for core data that are not stored in the Global Cache, such as when the originator specifies in the notification message that a certain dataset should not be cached. In these cases, the notification messages remain unchanged and point to the original source. Data consumers should subscribe to the notification messages from Global Caches instead of the notification messages from data providers for WMO core data. When data consumers receive a notification message, they should follow the URLs from that message, which either point to a Global Cache which has a copy of the data, or – in case of uncached content – point to the original source.

2.7.4.1 Technical considerations

- A Global Cache is built around three software components:
  - A highly available data server allowing data consumers to download cache resources with high bandwidth and low latency;
  - A Message Broker implementing both MQTTv3.1.1 and MQTTv5 to publish notification messages about resources that are available from the Global Cache;
  - A cache management system implementing the features needed to connect to the WIS ecosystem, receive data from WIS2 Nodes and other Global Caches, store the data on the data server and manage the content of the cache (expiration of data, deduplication, and so forth).
- A Global Cache will aim to contain copies of real-time and near real-time data designated as "core" within the WMO Unified Data Policy (Resolution 1 (Cg-Ext(2021))).
- A Global Cache will host data objects copied from NCs/DCPCs.
- A Global Cache will publish notification messages advertising the availability of the data objects it holds. The notification messages will follow the standard structure (see *Manual on WIS*, Volume II -Appendix E. WIS2 Notification Message).
- A Global Cache will use the standard topic structure in its local Message Brokers (see *Manual on WIS*, Volume II - Appendix D. WIS2 Topic Hierarchy).
- A Global Cache will publish to the topic `cache/a/wis2/…`.
- There will be multiple Global Cache to ensure the highly available, low-latency global provision

of real-time and near-real-time core data within WIS2.

- There will be multiple Global Caches that may attempt to download cacheable data objects from all originating centres with cacheable content. A Global Cache will also download data objects from other Global Caches. This will ensure that each Global Cache has full global coverage, even when direct download from an originating centre is not possible.
- Global Caches will operate independently of one another. Each Global Cache will hold a full copy of the cache – although there may be small differences between the various Global Caches as data availability notification messages propagate through WIS to each one. There is no formal synchronization between Global Caches.
- A Global Cache will temporarily cache all resources published on the `metadata` topic. A Global Discovery Catalogue will subscribe to notifications about the publication of new or updated metadata, download the metadata record from the Global Cache and insert it into the catalogue. A Global Discovery Catalogue will also publish a metadata record archive each day containing the complete content of the catalogue and advertise its availability with a notification message. This resource will also be cached by a Global Cache.
- A Global Cache is designed to support real-time content distribution. Data consumers access data objects from a Global Cache instance by resolving the URL in a data availability notification message and downloading the file to which the URL points. Only by checking the URL, is it transparent to the data consumers from which Global Cache they are downloading the data. There is no need to download the same data object from multiple Global Caches. The data id contained within notification messages is used by data consumers and Global Services to detect such duplicates.
- There is no requirement for a Global Cache to provide a browsable interface to the files in its repository in order to allow data consumers to discover what content is available. However, a Global Cache may choose to provide such a capability (for example, implemented as a WAF), along with documentation to inform data consumers of how the capability works.
- The default behaviour for a Global Cache is to cache all data published under the `origin/a/wis2/data/+/core` topic. A data publisher may indicate that data should not be cached by adding the `"cache": false` assertion in the WIS2 Notification Message.
- A Global Cache may decide not to cache data, for example, if the data are considered too large, or if a WIS2 Node publishes an excessive number of small files. If a Global Cache decides not to cache data, it should behave as though the cache property is set to false and shall send a WME (WIS2 Monitoring Event) message on the `monitor` topic hierarchy to inform other WIS centres about it. The Global Cache operator should work with the originating WIS2 Node and its GISC to remedy this issue.
- If core data are not cached on a Global Cache (that is, if the data are flagged as `"cache": false` or if the Global Cache decides not to cache these data), the Global Cache shall nevertheless republish the WIS2 Notification Message to the `cache/a/wis2/…` topic. In this case, the message id will be changed, and the rest of the message will not be modified.
- A Global Cache should operate with a fixed IP address so that WIS2 Nodes can permit access to download resources based on IP address filtering. A Global Cache should also operate with a publicly resolvable DNS name pointing to that IP address. The WMO Secretariat must be informed of the IP address and/or hostname, and any subsequent changes.
- A Global Cache should validate the integrity of the resources it caches and only accept data that

match the integrity value from the WIS2 Notification Message. If the WIS2 Notification Message does not contain an integrity value, the Global Cache should accept the data as valid. In this case, the Global Cache may add an integrity value to the message it republishes.

- As a convention, the Global Cache centre-id will be `{tld}-{centre-name}-global-cache`.

2.7.4.2 Practices and procedures

- A Global Cache shall subscribe to the topics `origin/a/wis2/#` and `cache/a/wis2/#`.
- A Global Cache shall ignore all messages received on the topics `origin/a/wis2/+/data/recommended/#` and `cache/a/wis2/+/data/recommended/#`[13]
- A Global Cache shall retain the data and metadata it receives for a minimum of 24 hours. Requirements relating to varying retention times for different types of data may be added later.
- For messages received on the topic `origin/a/+/data/core/#` or `cache/a/+/data/core/#`, a Global Cache shall:
  - If the message contains the property `"properties.cache": false`,
    - Republish the message at topic `cache/a/wis2/…`, matching `+/a/wis2/…` where the original message has been received, after having updated the id of the message.
  - Else
    - Maintain a list of `data_id` values that have already been downloaded;
    - Verify whether the message points to new or updated data by comparing the pubtime value of the notification message with the list of `data_id` values;
    - If the message is new or updated:
      - Download only new or updated data from the `href` or extract the data from the message content;
      - If the message contains an integrity value for the data, verify the integrity of the data;
      - If data is downloaded successfully, move the data to the HTTP endpoint of the Global Cache;
      - Wait until the data becomes available at the endpoint;
      - Add the property `properties.global-cache` with the value of the centre identifier of the Global Cache to identify the origin of the data source downloaded by downstream applications;
      - Modify the message identifier and the canonical link's `href` of the received message and leave all other fields untouched;
      - Republish the modified message to topic `cache/a/wis2/…` ,matching the `+/a/wis2/…` where the original message has been received;
      - The metric `wmo_wis2_gc_downloaded_total` will be increased by 1; The metric `wmo_wis2_gc_dataserver_last_download_timestamp_seconds` will be updated with the timestamp (in seconds) of the last successful download from the WIS2 Node or Global Cache;
    - Else

        - Drop the messages for data already present in the Global Cache.
- If the Global Cache is not able to download the data, the metric `wmo_wis2_gc_downloaded_error_total` will be increased by 1.
- A Global Cache shall provide the metric defined in this Guide at an HTTP endpoint.
- A Global Cache should make sure that data are downloaded in parallel and that downloads are not blocking each other.
- A Global Cache will implement Status Code Definitions as defined by the HTTP specification.
- The metric `wmo_wis2_gc_dataserver_status_flag` will reflect the status of the connection to the download endpoint of the centre. Its value will be 1 when the endpoint is up and 0 otherwise.
- The metric `wmo_wis2_gc_last_metadata_timestamp_seconds` will reflect the datetime (as a timestamp, the number of seconds since the UNIX epoch) of the last metadata resource processed by a given centre.

2.7.5 Global Discovery Catalogue

2.7.5.1 Technical considerations

- The Global Discovery Catalogue provides data consumers with a mechanism for discovering and searching for datasets of interest as well as learning how to interact with and find out more information about those datasets.
- The Global Discovery Catalogue implements the OGC API – Records – Part 1: Core standard[14], adhering to the following conformance classes and their dependencies:
    - Searchable Catalog (Deployment);
    - Searchable Catalog - Sorting (Deployment);
    - Searchable Catalog - Filtering (Deployment);
    - JSON (Common Component);
    - HTML (Common Component).
- The Global Discovery Catalogue will make discovery metadata available via the collection identifier `wis2-discovery-metadata`.
- The Global Discovery Catalogue advertises the availability of datasets and how to access them or subscribe to updates.
- The Global Discovery Catalogue does not advertise or list the availability of individual data objects that comprise a dataset (that is, data files).
- A single Global Discovery Catalogue is sufficient for WIS2.
- Multiple Global Discovery Catalogues may be deployed for resilience.
- Global Discovery Catalogues operate independently of each other; each Global Discovery Catalogue holds all discovery metadata records. Global Discovery Catalogues do not need to synchronize with each other.
- A Global Discovery Catalogue is populated with discovery metadata records from a Global Cache and receives messages about the availability of discovery metadata records via a Global Broker.

- The subscription topic shall be `cache/a/wis2/+/metadata/#`.

- A Global Discovery Catalogue should connect to and subscribe to more than one Global Broker to ensure that no messages are lost in the event of a Global Broker failure. A Global Discovery Catalogue will discard duplicate messages as needed.
- A Global Discovery Catalogue will verify that a discovery metadata record identifier's centre-id token (see Manual on WIS, Volume II – Appendix F. WMO Core Metadata Profile (Version 2)) matches the centre-id level of the topic from which it was published (see Manual on WIS, Volume II – Appendix D. WIS2 Topic Hierarchy) to ensure that discovery metadata are published by the authoritative organization.
- A Global Discovery Catalogue will validate discovery metadata records against the WCMP2. Valid WCMP2 records will be ingested into the catalogue. Invalid or malformed records will be discarded and reported to the Global Monitor against the centre-id associated with the discovery metadata record.
- A Global Discovery Catalogue may perform quality assessment of discovery metadata against the WCMP2 Key Performance Indicators (KPIs) [15], providing the following metrics as defined in the WIS2 Metric Hierarchy [16]: `wmo_wis2_gdc_kpi_percentage_total`, `wmo_wis2_gdc_kpi_percentage_average`, `wmo_wis2_gdc_kpi_percentage_over80_total`.
- A single Global Discovery Catalogue perfoming quality assessment is sufficient for WIS2.
- A Global Discovery Catalogue will only update discovery metadata records to replace links for dataset subscription and notification (origin), with their equivalent links for subscription at Global Brokers (cache).
- A Global Discovery Catalogue will periodically assess discovery metadata provided by NCs and DCPCs against a set of key performance indicators (KPIs) in support of continuous improvement. Suggestions for improvement will be reported to the Global Monitor against the centre identifier associated with the discovery metadata record.
- A Global Discovery Catalogue will remove discovery metadata that are marked for deletion as specified in the data notification message.
- A Global Discovery Catalogue should apply faceting capability as specified in the cataloguing considerations of the WCMP2 specification, as defined in OGC API - Records.
- A Global Discovery Catalogue will provide human-readable web pages with embedded markup using the schema.org vocabulary, thereby enabling search engines to crawl and index its content. Consequently, data consumers should be able to discover WIS content via third party search engines.
- A Global Discovery Catalogue will generate and store a zip file of all WCMP2 records once a day; this file will be made be accessible via HTTP. The zipfile will include a directory named after the centre-id of the Global Discovery Catalogue containing all WCMP2 records.
- A Global Discovery Catalogue will publish a WIS2 Notification Message of its zip file of all WCMP2 records on its centre-id's metadata topic (for example, `origin/a/wis2/centre-id/metadata`, where `centre-id` is the centre identifier of the Global Discovery Catalogue).
- A Global Discovery Catalogue may initialize itself (cold start) from a zip file of all WCMP2 records published.
- As a convention, a Global Discovery Catalogue's centre-id will be `{tld}-{centre-name}-global-`

`discovery-catalogue`.

2.7.5.2 Global Discovery Catalogue Reference Implementation: wis2-gdc

To provide a Global Discovery Catalogue, Members may use whichever software components they consider most appropriate to comply with the WIS2 technical regulations.

To assist Members in participating in WIS2, a free and open-source Global Discovery Catalogue Reference Implementation, wis2-gdc, is available for download and use. wis2-gdc builds on mature and robust free and open-source software components that are widely adopted for operational use.

wis2-gdc provides the functionality required for the Global Discovery Catalogue, including the following technical functions:

- Discovery metadata subscription and publication from the Global Broker;
- Discovery metadata download from the Global Cache;
- Discovery metadata validation, ingest and publication;
- WCMP2 compliance;
- Quality assessment (KPIs);
- OGC API - Records - Part 1: Core compliance;
- Metrics reporting;
- Implementation of metrics.

wis2-gdc is managed as a free and open source project. Source code, issue tracking and discussions are hosted openly on GitHub: https://github.com/wmo-im/wis2-gdc.

2.7.6 Global Monitor

2.7.6.1 Technical considerations

- WIS standardizes how system performance and data availability metrics are published from WIS2 Nodes and Global Services.
- For each type of Global Service, a set of standard metrics has been defined. Global Services will implement and provide an endpoint for those metrics to be scraped by the Global Monitor.
- The Global Monitor will collect metrics as defined in the OpenMetrics standard.
- The Global Monitor will monitor the "health" (that is, the performance) of components at NCs/DCPCs, as well as Global Services.
- The Global Monitor will provide a web-based dashboard that displays the WIS2 system performance and data availability.
- As a convention, the Global Monitor centre-id will be `{tld}-{centre-name}-global-monitor`.
- The main task of the Global Monitor will be to regularly query the metrics provided by the relevant WIS2 entities, aggregate and process the data and then provide the results to the end user in a suitable presentation.

2.8 Operations

2.8.1 Interoperability with external systems

Driven by international standards for data discovery, access, and visualization, the WIS2 principles help lower the barrier for WMO Members to weather, climate, and water data. An additional benefit of adopting these standards is that Members are able to provide the same data and access mechanisms to external systems at no extra cost for implementation.

WIS2 standards are based on industry standards (OGC, W3C, IETF) and allow for broad interoperability. This means that non-traditional users can also use data from WIS2 data in the same manner, without the requirement for specialized software, tools, or applications.

2.8.1.1 Publishing meteorological data through WIS2 into ICAO SWIM

Meteorological data is an essential input for public weather services and aviation services alike. WIS2 provides the mechanism for data exchange in WMO, while SWIM is the ICAO initiative to harmonize the provision of aeronautical, meteorological and flight information to support air traffic management (ATM).

Both WIS2 and SWIM support similar outcomes regarding data exchange. However, there are differences with respect to both approach and implementation.

Specifications for WIS2 are defined in the *Manual on WIS*, Volume II, and further elaborated in this Guide. Specifications for SWIM will be defined in the Procedures for Air Navigation Services –Information Management (PANS-IM) (ICAO Doc. 10199).[17]

During the WIS2 transition phase (2025-2033), meteorological data published via WIS2 will automatically be published to GTS via the WIS2-to-GTS gateways.

WIS2	**SWIM**
Earth system scope: Weather, climate, hydrology, atmospheric composition, cryosphere, ocean and space weather data.	ATM scope: Aeronautical, meteorological and flight information.
Data centric: A consumer discovers data and then determines the services through which those data may be accessed.	Service centric: A consumer discovers a service (or service provider) and determines what resources (that is, information) are available therein.
Technical protocols: MQTT, HTTP	Technical protocols: AMQP[18]

An organization (for example, the National Meteorological Service) that is responsible for providing meteorological data to WIS2 may be designated by the ICAO Contracting State as a responsible entity for providing aeronautical meteorological information into SWIM. Where requirements dictate, the organization may provide regional capability on behalf of a group of countries or territories.

This section of the Guide outlines how such an organization may efficiently provide the required data/information to the two systems. It proposes an interoperability approach between WIS2 and

SWIM where meteorological data published via WIS2 can be automatically propagated to SWIM.

This Guide covers only how data from WIS2 can be published into SWIM. It does not address the consumption of information from SWIM services.

It also does not cover details regarding the implementation of the SWIM services - including, but not limited to:

- Mechanisms used by SWIM to discover service providers and services;
- Specifications of SWIM data messages;
- AMQP Message Broker configuration;
- Operation, logging and monitoring;
- Cybersecurity considerations for the provision of SWIM services.

This Guide will be updated as more information is made available from ICAO and/or as recommended practices are updated.

Finally, it should also be noted that the provision of aeronautical meteorological information and its exchange via the ICAO Aeronautical Fixed Service (AFS) are defined solely under the ICAO regulatory framework and are therefore beyond the scope of this Guide.

2.8.1.1.1 WIS2 to SWIM gateway

The WIS2 to SWIM interoperability approach employs a gateway component (as per Figure 2):

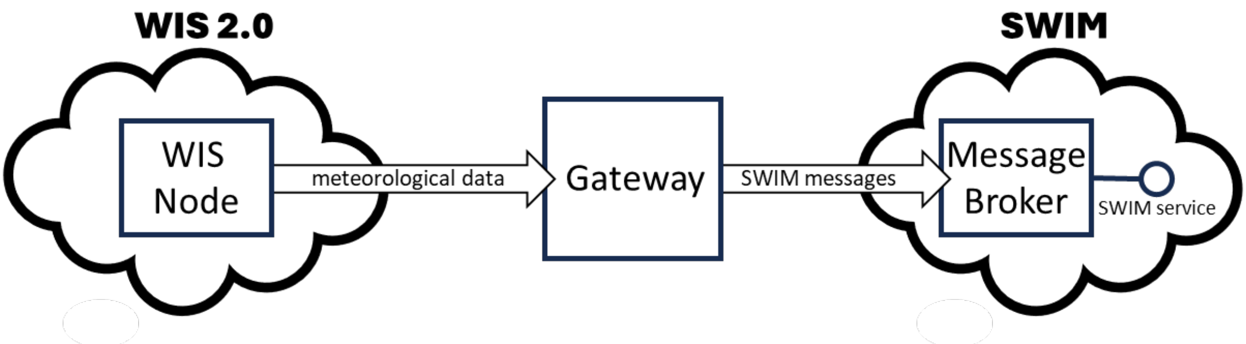

Figure 2. Schematic of an interoperability approach

The gateway component can operate as an "adapter" between WIS2 and SWIM by pulling the requisite meteorological data from WIS2 and re-publishing it to SWIM.

2.8.1.1.2 Data types and format

Specifications for aeronautical meteorological information are provided in ICAO Annex 3 and other relevant guidance materials. The ICAO Meteorological Information Exchange Model (IWXXM) format (FM 205)[19] is to be used for encoding aeronautical meteorological information in SWIM.

2.8.1.1.3 Publishing meteorological data via WIS2

For meteorological data to be published from WIS2 to SWIM, the organization responsible for providing the data needs to operate a WIS2 Node and comply with the pertinent technical regulations specified in the *Manual on WIS*, Volume II. Onward distribution of the data by the Message Broker over SWIM can be handled by the respective Information Service Provider in accordance with ICAO Standards and Recommended Practices (SARPs).

The responsible organization should consider whether these data should be published via an existing WIS2 Node, or whether a separate WIS2 Node should be established. For example, the data may be provided by a separate operational unit, or there may be a requirement to easily distinguish between data for SWIM and other meteorological data.

If a new WIS2 Node is needed, the responsible organization must establish one and register it with the WMO Secretariat. For more information, see 2.6 Implementation and operation of a WIS2 Node.

Datasets are a central concept in WIS2. Where meteorological data is published via WIS2, it will be packaged into datasets. The data should be grouped at the country/territory level (for instance, datasets should be published for a given country/territory), one for each datatype (for example, aerodrome observation, aerodrome forecast, quantitative volcanic ash concentration information, and so forth).

For the purposes of publishing through WIS2, datasets containing aeronautical meteorological information should be considered as recommended data, as described in Resolution 1 (Cg-Ext(2021)). The recommended data category of the policy is intended to cover data that should be exchanged by Members to support Earth system monitoring and prediction efforts.

Recommended data:

- May be subject to conditions on use and reuse;
- May have access controls[20][21] applied at the WIS2 Node;
- Are not cached within WIS2 by the Global Caches.[22]

Resolution 1 (Cg-Ext(2021)) requires transparency on the conditions of use for recommended data. Conditions regarding the use of aeronautical meteorological information are specified in ICAO Annex 3 and, optionally, by the ICAO Contracting State. Such conditions of use should be explicitly stated in the discovery metadata for each dataset, as described below.

- The attribute `wmo:dataPolicy` should be set to `recommended`.
- Information about conditions of use should be specified using the `rights` property (see the example below) and/or a `link` object with a relation `license`.
- Information about access control should be specified using a `security` object in the `link` object describing the data access details.

The following is an example expression of conditions relating to the use of aeronautical meteorological information:

```
"properties": {
  ...
  "rights": "This information is freely disseminated for the purposes of safety of
international air navigation. ICAO Annex 3"
  ...
```

```
}
```

For more information on the WMO Core Metadata Profile, see the *Manual on WIS*, Volume II, Appendix F. WMO Core Metadata Profile (Version 2).

On receipt of new data, the WIS2 Node will:

1. Publish the data as a resource via a web server (or web service);
2. Publish a WIS2 Notification Message advertising the availability of the data resource to a local Message Broker.

Note that, in contrast to GTS, WIS2 publishes data resources individually, each with an associated notification message. WIS2 does not group data resources into bulletins.

The data resource is identified using a URL. The notification message refers to the data resource using this URL.[23]

For more details on the WIS2 Notification Message, see the *Manual on WIS*, Volume II, Appendix E. WIS2 Notification Message.

The notification message must be published to the proper topic on the Message Broker. WIS2 defines a standard topic hierarchy to ensure that data are published consistently by all WIS2 Nodes. Notification messages for aviation data should be published on a specific topic allowing a data consumer, such as the gateway, to subscribe only to aviation-specific notifications. See the example below.

Example topic used to publish notifications about quantitative volcanic ash concentration information:

```
origin/a/wis2/{centre-id}/data/recommended/weather/aviation/qvaci
```

For more details on the WIS Topic Hierarchy, see the *Manual on WIS*, Volume II, Appendix D. WIS2 Topic Hierarchy.

WIS Global Brokers subscribe to the local Message Brokers of WIS2 Nodes and republish notification messages for global distribution.

At a minimum, the WIS2 Node should retain the aviation data for a duration that meets the needs of the gateway. A retention period of at least 24 hours is recommended.

2.8.1.1.4 Gateway implementation

The potential interactions between the gateway component, WIS2 and SWIM are illustrated in Figure 3.[24]

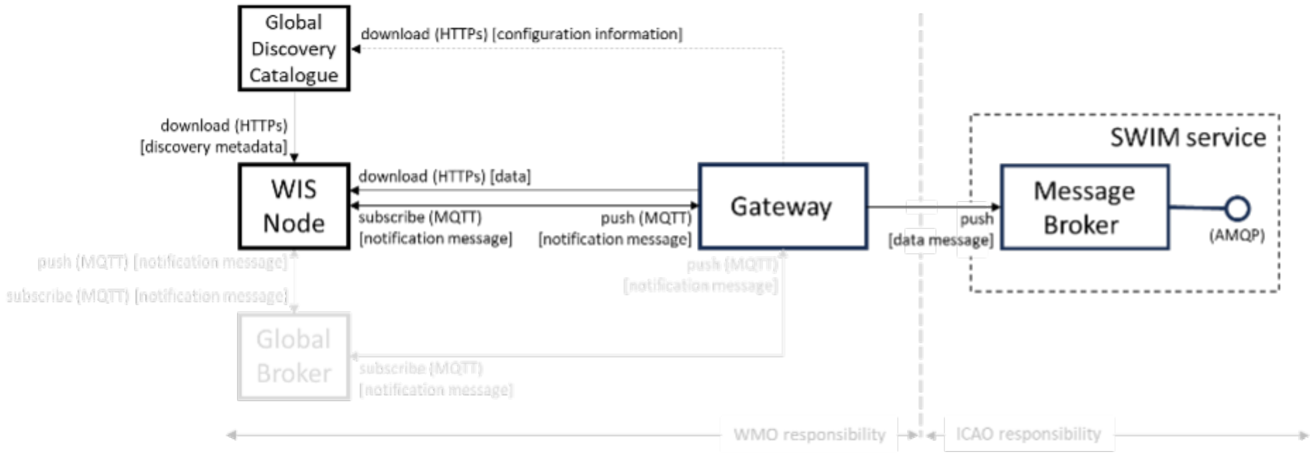

Figure 3. Interactions between the gateway component and WIS2 and SWIM components

Configuration

Dataset discovery metadata will provide useful information that can be used to configure the gateway component, for example, the topic(s) to subscribe to plus additional information that may be needed for the SWIM service.

Discovery metadata can be downloaded from the Global Discovery Catalogue.

Functions

The gateway component implements the following functions:

- Subscribe to the pertinent topic(s) for notifications about new aeronautical meteorological information;[25]
- On receipt of notification messages about newly available data:
  - Parse the notification message, discarding duplicate messages already processed;
  - Download the data resource from the WIS2 Node[26] using the URL in the message - the resource should be in IWXXM format;
  - Create a new data message as per the SWIM specifications, including the unique identifier extracted from the data resource[27], and embed the aviation weather data resource within the data message;
  - Publish the data message to the appropriate topic on the SWIM Message Broker component of the SWIM service.

The choice of protocol for publishing to the SWIM Message Broker should be based on a bilateral agreement between operators of the gateway component and the SWIM service.

The gateway component should implement logging and error handling as necessary to enable reliable operations. WIS2 uses the OpenMetrics standard[28] to publish metrics and other operating information. The use of OpenMetrics by the gateway component would enable monitoring and performance reporting to be easily integrated into the WIS2 system.

Operation

The gateway component may be operated at the national or regional level depending on the organizational governance in place.

2.8.1.1.5 SWIM service

The SWIM aviation weather information service may include of a Message Broker component which implements the Advanced Message Queuing Protocol (AMQP) 1.0 messaging standard.[29]

The Message Broker publishes the data messages provided by the gateway component.

The Message Broker must ensure that data messages are provided only by authorized sources, such as a gateway, and should validate incoming messages as aeronautical meteorological information.

2.8.1.2 Ocean Data and Information System

The Ocean Data and Information System (ODIS) is a federation of independent data systems, coordinated by the International Oceanographic Data and Information Exchange (IODE) of the Intergovernmental Oceanographic Commission of the United Nations Educational, Scientific and Cultural Organization (IOC-UNESCO). This federation includes continental-scale data systems, as well as those of small organizations. ODIS partners use web architectural approaches to share metadata describing their holdings, services, and other capacities. In brief, IODE publishes guidelines on how to share metadata as linked open data, serialized in JSON-LD using schema.org[30] semantics. ODIS nodes use these guidelines to publish their metadata catalogues on the Web. This allows all systems with web connectivity to harvest and merge these catalogues, creating a global map of the ocean data. IODE harvests all metadata shared by ODIS partners, combines these metadata and creates a knowledge graph, and processes these metadata to export derivative products (for example, diagnostic reports and cloud-optimized data products). The Ocean InfoHub (OIH) system is IODE's Reference Implementation of a discovery system leveraging ODIS. ODIS architecture and tools are free and open-source software (FOSS), with regular releases published for the community.

To reach beyond the oceans domain, ODIS works with other data systems and federations, such as WIS2, to define sustainable data and metadata exchanges and - where needed - translators or converters. The resources needed to convert between such systems are developed in the open and in close collaboration with staff from those systems. These exchanges include the extract, transform and load (ETL) functions to ensure that the bilateral exchange is mutually beneficial.

2.8.1.2.1 Cross system interoperability

Given the strong support for standards and interoperability by both WIS2 and ODIS, data and metadata exchange are carried out using web architecture principles and approaches. The ability to discover ODIS data on WIS2 (and the reverse) is a goal in extending the reach of both systems and data beyond their primary communities of interest.

WIS2 Global Discovery Catalogues will provide discovery metadata records using the OGC API - Records standard. The Global Discovery Catalogues will include schema.org and JSON-LD annotations on WCMP2 discovery metadata to enable cross-pollination and federation.

ODIS dataset records will be made available using the WCMP2 standard and provided as objects available via HTTP for ingest, validation and publication to the Global Discovery Catalogues as a federated catalogue. ODIS data will be published as recommended data as per the WMO Unified

Data Policy (Resolution 1 (Cg-Ext(2021))). (See Figure 4)

[Figure 4. WIS2 and ODIS metadata and catalogue interoperability] | *images/wis2-odis-metadata-discovery-interop.png*

Figure 4. WIS2 and ODIS metadata and catalogue interoperability

As a result, federated discovery will be realized between both systems, users will be able to access the data from as close as possible to their source, and the data will be able to be used and reused in an authoritative manner.

[1] See MQTT Specifications: https://mqtt.org/mqtt-specification/.

[2] See RFC 7231 - Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content: https://datatracker.ietf.org/doc/html/rfc7231.

[3] See IANA TLDs: https://data.iana.org/TLD.

[4] See http://www.metoffice.gov.uk.

[5] The ".gov" part of the domain name is superfluous for the purposes of WIS2 There is nothing preventing its use, but it does not add any value.

[6] The default connection credentials for a WIS2 Node Message Broker are username `everyone` and password `everyone` WIS2 Node operators should choose credentials that meet their local policies (for example, password complexity).

[7] In WIS2, IP addresses are used to determine the origin of connections and confer trust to remote systems. It is well documented that IP addresses can be hijacked and that more sophisticated mechanisms, such as Public Key Infrastructure (PKI), are available for reliably determining the origin of connection requests. However, the complexities of implementing such mechanisms create barriers to Member participation in WIS2. For the purposes of WIS2, which involves distributing publicly accessible data and messages, IP addresses are considered to provide an adequate level of trust.

[8] In some cases, WIS2 Nodes will need to serve core data directly (see 1.3.3.5 Considerations when providing core data in WIS2). In these situations, the WIS2 Node data server must remain publicly accessible.

[9] See OpenAPI Specification - Security Scheme Object: https://spec.openapis.org/oas/v3.1.0#security-scheme-object.

[10] See OpenMetrics: https://openmetrics.io.

[11] OpenMetrics is proposed as a draft standard within the Internet Engineering Task Force (IETF).

[12] See https://github.com/wmo-im/wis2-metric-hierarchy.

[13] It is also technically possible to filter recommended data by using a wildcard subscriptions such as `origin/a/wis2/+/data/core/#`. However, avoiding wildcard subscription is generally considered good practice as it limits the burden of the broker operated by Global Brokers.

[14] See OGC-API Records - Part 1 https://docs.ogc.org/is/20-004r1/20-004r1.html.

[15] https://wmo-im.github.io/wcmp2/kpi/wcmp2-kpi-STABLE.html

[16] https://github.com/wmo-im/wis2-metric-hierarchy/blob/main/metrics/gdc.csv

[17] PANS-IM is expected to available on ICAO NET by July 2024 and to become applicable in November 2024. The information provided herein is based on draft proposals from ICAO.

[18] AMQP 1.0 is one of the protocols proposed in the draft PANS-IM.

[19] IWXXM (FM 205) is defined in the *Manual on Codes* (WMO-No. 306), Volume I.3.

[20] WIS2 follows OpenAPI recommendations regarding the security schemes for authenticated access - either HTTP authentication, API keys, OAuth2 or OpenID Connect Discovery. For more information, see OpenAPI Security Scheme Object: https://spec.openapis.org/oas/v3.1.0#security-scheme-object.

[21] WIS2 does not provide any guidance on use of Public Key Infrastructure (PKI).

[22] Global Caches enable the highly available, low-latency distribution of core data. Given that core data is provided on a free and unrestricted basis, Global Caches do not implement any data access control.

[23] Where the data resource does not exceed 4 Kb, it may be embedded in the notification message.

[24] Note that the figure simplifies the transmission of discovery metadata from the WIS2 Node to the Global Discovery Catalogue. The WIS2 Node publishes notification messages advertising the availability of a new discovery metadata resource at a given URL. These messages are republished by the Global Broker. The Global Discovery Catalogue subscribes to a Global Broker and downloads the discovery metadata from the WIS2 Node using the URL supplied in the message.

[25] WIS2 recommends subscribing to notifications from a Global Broker. However, where both a gateway component and a WIS2 Node are operated by the same organization, it may be advantageous to subscribe directly to the local Message Broker of WIS2 Node, for example, to reduce latency.

[26] The WIS2 Node may control access to data. If this is the case, the gateway component will need to be implemented accordingly

[27] In case a unique identifier is required for proper passing of an aviation weather message to the gateway component, the GTS abbreviated heading (TTAAii CCCC YYGGgg) in the COLLECT envelop can be used (available in IWXXM messages that have a corresponding TAC message). Alternatively, content in the attribute `gml:identifier` (available in newer IWXXM messages such as WAFS SIGWX Forecast and QVACI), may also serve this purpose. There is currently no agreed definition for a unique identifier for IWXXM METAR and TAF reports for individual aerodromes.

[28] See OpenMetrics: https://openmetrics.io.

[29] See AMQP 1.0: https://www.amqp.org/resources/specifications.

[30] See https://schema.org.

PART III. Additional components

3.1 Overview

In addition to the core set of Global Services defined, additional Global Services may exists in WIS2.

Additional Global Services:

- provide value added capabilities and insights into WIS2
- are not required components of WIS2
- may be provided with various access control policies depending on their scope

3.2 Other services

3.2.1 Global Replay

A Global Replay provides access and subscription to past WIS Notification Messages.

3.2.1.1 Component definition

- WIS2 may include a Global Replay.
- The Global Replay provides the Global Cache, Global Discovery Catalogue and data consumers with a mechanism to search, query and subscribe to past notification messages and monitoring events of interest as published by the Global Broker.
- A Global Replay subscribes to all notification messages and monitoring events as published by Global Broker. It stores the notification message and monitoring events.
- A Global Replay shall store copies of notification messages and monitoring events for a duration compatible with the real-time or near real-time schedule of the data and not less than:
  - 1 day for notification messages.
  - 7 days for monitoring events.
- A Global Replay will discard notification messages and monitoring events once the retention period has expired.

3.2.1.2 Technical considerations

- The Global Replay provides Global Services and data consumers with a mechanism to search, query and subscribe to notification messages and monitoring events of interest.
- The Global Replay implements the OGC API – Features – Part 1: Core standard[1], adhering to the following conformance classes and their dependencies:
  - Core
  - GeoJSON
- The Global Replay implements the OGC API – Processes - Part 1: Core standard[2], adhering to the following conformance classes and their dependencies:

    - Core
    - OGC Process Description
    - JSON
- A Global Replay shall subscribe to all Global Broker instances, to the topics `origin/a/wis2/#`, `cache/a/wis2/#` and `monitor/a/wis2/#`.
- A Global Replay shall use the notification message and monitoring event identifier to deduplicate notifications and events.
- The Global Replay will make notification messages available via the API collection identifier `wis2-notification-messages`.
- The Global Replay will make monitoring events available via the API collection identifier `wis2-monitoring-events`.
- The Global Replay will provide an API process to trigger the republication of previously sent messages, allowing users to subscribe to messages with various criteria (spatial and / or temporal extent, topic selection) via the Global Broker.
    - the API process request accepts the following inputs:
        - `datetime`: a temporal range of messages to republish (example `2025-04-06T17:21:26Z/2025-04-06T19:21:26Z`).
        - `subscriber-id`: a unique identifier (UUID) defined by the user/client. This identifier may be reused.
        - `topic`: the topic on which to trigger message replication (centre-id required, allowing `+` and `#` for subsequent topic levels)
    - the resulting topic subscription in the API process response will be `replay/a/wis2/<centre-id>/<subscriber-id>/<topic>`, where:
        - `topic` is the full topic of the message as previously published to WIS2.
    - the API process will set temporal boundary limits to 15 minutes. When exceeded, it will result in an error message prohibiting the republishing of messages.
    - an example topic would be `replay/a/wis2/ca-eccc-msc-global-replay/fdb6dae4-95fe-465a-80fb-498ebd61dce8/cache/a/wis2/br-inmet/data/core/weather/surface-based-observations/synop`
- A single Global Replay instance is sufficient for WIS2.
- Multiple Global Replay instances may be deployed for resilience.
- Global Replay instances operate independently of each other; each Global Replay instance will hold notification messages and monitoring events according to the required retention period. Global Replays do not need to synchronize with each other.
- A Global Replay is populated with notification messages and monitoring events received via a Global Broker instance.
- A Global Replay is connected to at least two (2) Global Brokers.
- A Global Replay instance will discard duplicate messages as needed.
- A Global Replay can validate notification messages against the standard format (see *Manual on*

WIS, Volume II – Appendix E. WIS2 Notification Message). Valid messages will be stored. Invalid or malformed messages will be discarded.

- A Global Replay can validate monitoring events against the WIS Monitoring Events Message Encoding (see *Manual on WIS*, Volume II – Appendix TODO. WIS2 Monitoring Events Message Encoding). Valid WIS Monitoring Events Message Encoding Messages will be stored. Invalid or malformed monitoring events will be discarded.
- A Global Replay will remove notification messages after the required retention period.
- A Global Replay will remove monitoring events after the required retention period.
- A Global Replay may choose to implement access control for controlling usage.
- As a convention Global Replay centre-id will be `tld-{centre-name}-global-replay`.

3.2.1.3 Global Replay Reference Implementation: wis2-grep

To provide a Global Replay, members may use whichever software components they consider most appropriate to comply with WIS2 Technical Regulations.

To assist Members participation in WIS2, a free and open-source Global Replay Reference Implementation is made available for download and use. wis2-grep builds on mature and robust free and open-source software components that are widely adopted for operational use.

wis2-grep provides functionality required for the Global Replay, providing the following technical functions:

- notification messages and monitoring event subscription from a Global Broker insrtance
- notification message and monitoring event ingest and API publication
- user defined subscription for MQTT message replay from a Global Broker instance
- OGC API - Features - Part 1: Core compliance
- OGC API - Processes - Part 1: Core compliance

wis2-grep is managed as a free and open source project. Source code, issue tracking and discussions are hosted in the open on GitHub: https://github.com/wmo-im/wis2-grep.

3.3 Sensor Centres

Sensor Centres are components external to WIS 2.0, which provide continuous assessments on functionality. Sensor Centres can provide external feedback on WIS 2.0 functions to trigger corrective action in support of continuous improvement.

3.3.1 Overview

As described in the Manual on WIS, Volume II, WIS 2.0 is a collective, distributed data exchange system that requires monitoring.

All WIS 2.0 components have defined metrics and conditions on those metrics to trigger alerts.

The Global Monitor collects the metrics and presents them a suite of dashboards enabling observability, visuaization and analysis.

Beyond WIS 2.0 itself, a Sensor Centre relies on WIS 2.0 Global Services and Nodes. It implements further processing on any component of the WIS 2.0 architecture. This processing results in additional metrics which can also be provided to dashboards, within or external to WIS 2.0.

"Sensor Centre" is a generic term to describe additional components to measure various aspects including (but not limited to):

- the quality and performance of Global Services
- the conformance of the products exchanged with the agreed WMO Regulations (eg. RBON)

Each type of Sensor Centre will result in a set of metrics that will be used to measure and evaluate WIS 2.0 Global Services and Nodes.

A Sensor Centre may also provide logs, in addition to the defined metrics, through download links announced using the WIS2 Monitoring Events standard in support of further analysis and corrective action.

Sensor Centres monitoring the Global Services that are critical to the function of WIS 2.0 are described below: Sensor Centre Global Broker, Sensor Centre Global Cache, and Sensor Centre Global Discovery Catalogue.

Given that Sensors Centres may be used to monitor any function in WIS 2.0, further Sensor Centres may be proposed to meet additional requirements, such as validation of GBON or RBON compliance. Every Sensor Centre must follow the standard mechanism for publishing metrics for presentation in dashboards that can be used to monitor the performance of the WIS 2.0 function being assessed.

Any Member may request to operate a Sensor Centre via the designation process - either for a new or existing type of Sensor Centre. Requests must specify what type of Sensor Centre will be operated and refer to the specification they will implement.

The specification of a Sensor Centre must be published in a publicly accessible location, such as a public GitHub repository. Where a Sensor Centre has a role in the objective assessment of Global Service performance, the Sensor Centre specification is subject to approval by INFCOM and will be published in the Guide to WIS, Vol 2, Part III. Where a Sensor Centre has some other official role (e.g., in support of RBON compliance) the responsible body for that function may assert their own governance requirements.

Where a Sensor Centre provides insight about the performance of WIS 2.0 as a whole, Global Monitors may agree to collect metrics from the Sensor Centre and provide monitoring dashboards. As part of the designation process, a Member may include a request for Global Monitors to harvest metrics, including suggested thresholds for metrics that would trigger a Global Monitor to publish an alert or warning.

3.3.2 Sensor Centres for Global Brokers (SCGB)

The aim of the SCGB is to compare the behaviour of Global Broker(s)

Through direct subscription to WIS2 Nodes or via other Global Brokers, each Global Broker aims at providing all messages available on WIS 2.0 to users subscribing to their MQTT(S) endpoint.

Therefore, the primary function of a SCGB is to test and evaluate that all GBs in WIS 2.0 are making available the same number of messages. In addition, it will provide a connectivity flag to detect the status of the connection toward each GB, as well as a timestamp of the last message being received from each GB.

As a convention, the Sensor Centre for Global Broker centre-id will be `{tld}-{centre-name}-sensor-centre-global-broker`.

For such a centre operated by Direction de la Météorologie of Congo, the centre-id would be cg-dirmet-sensor-centre-global-cache

3.3.2.1 Technical considerations

- The SCGB will connect to all Global Brokers.
- The SCGB subscribes to `origin/a/wis2/#`, `cache/a/wis2/#`, `monitor/a/wis2/#` on all GBs
- When the SCGB is successfully connected to the GB, the value of `wmo_wis2_scgb_connected_flag` will be set to `1`. When the SCGB cannot connect to the GB, the value of `wmo_wis2_scgb_connected_flag` will be set to `0`.
- When a message is received on a subscription, the SCGB will increment the metric `wmo_wis2_scgb_messages_received_total` by `1` and will set the metric `wmo_wis2_scgb_last_message_timestamp_seconds` to the current timestamp in seconds.

All the metrics must be exposed for scraping by the Global Monitor.

3.3.3 Sensor Centres for Global Caches (SCGC)

The aim of the SCGC is to compare the behaviour of Global Cache(s)

In WIS 2.0 each Global Cache is independent of the other Global Caches. According to the specification of WIS 2.0, Global Cache (see https://wmo-im.github.io/wis2-guide/guide/wis2-guide-APPROVED.html#_2_7_4_1_technical_considerations) :

```
Global Caches will operate independently of one another. Each Global Cache will hold a full copy of the cache  although there may be small differences between the various Global Caches as data availability notification messages propagate through WIS to each one. There is no formal synchronization between Global Caches.
```

it is therefore interesting to verify, for example:

1. what is the average delay to cache the data made available by WIS2 Node
2. are all files published as *core data* (and with `cache: true` in the Notification Message) by WIS2 Nodes are effectively available in all Global Caches
3. are files missed by some Global Cache or more critically by all Global Caches

Such metrics would provide useful information to identify problems, to help Global Caches to fix them, to define KPI that could be used to objectively measure the effective performance of the Cache.

It could also be used to detect anomalies from the WIS2 node, such as the reuse too frequently of the same `data_id`.

As a convention, the Sensor Centre for Global Cache centre-id will be `{tld}-{centre-name}-sensor-centre-global-cache`.

For such a centre operated by Météo-France the centre-id would be fr-meteofrance-sensor-centre-global-cache

3.3.3.1 Technical considerations

A list of metrics has been defined for this Sensor Centre at https://github.com/wmo-im/wis2-metric-hierarchy/blob/main/metrics/scgc.csv.

The processing of this Sensor Centre is as follows:

- Subscribe to a given `origin/a/wis2/…`, it could be `#` or a particular centre-id on at least one Global Broker
- Subscribe to a given `cache/a/wis2/…`, it could be `#` or a particular centre-id on at least one Global Cache - The subscription must be done on the broker of the Global Cache (unlike normal subscription to be made only on the Global Broker)

Only on the subscription to the Global Broker:

- Discard the Notification Message if is it `recommended` data as it will not be cached
- Detect any duplicates `data_id` not including a `rel: update` within a period of at least 3 hours
- Increase by 1 `wmo_wis2_scgc_messages_published_total` metrics
- Store the time where the message as been received
- (optional) Store the full Notification Message - this can be useful to analyse systematic issues

For each of the subscription to the various Global Caches:

- For each Notification Message and using `data_id`, make the difference between the time was received on `origin` and the same `data_id` on `cache`: $\text{Time}_{\text{Cache}}$ - $\text{Time}_{\text{Origin}}$
- Update the `wmo_wis2_scgc_cache_delay_seconds` metric with this value
- Compare this value with the three threshold defined in the metric table above. Increase by 1 `wmo_wis2_scgc_messages_cached_delay_total` using the threshold as a label (so less than 120s, less that 300s, less than 600s)
- If no Notification Message for the `data_id` is received after the highest threshold (here 600s), increase by 1 `wmo_wis2_scgc_messages_missed_total`

As a Global Cache can decide not to cache a data granule, and in order to assess if a Global Cache has decided to do so, the Sensor Centre will compare the `links` array published on `origin/a/wis/#` and in the Notification Message received on `cache/a/wis2/#` if the two `links` are identical, then the

Global Cache has decided not to cache the data. In this case, increase by 1 `wmo_wis2_scgc_messages_cache_override_total`.

If no Global Cache has cached the data, increase by 1 `wmo_wis2_scgc_messages_missed_all_total`

All the metrics must be exposed for scraping by the Global Monitor.

3.3.4 Sensor Centres for Global Discovery Catalogues (SCGDC)

The primary function of an SCGDC is to test and evaluate content parity across all GDCs in WIS 2.0.

A Global Discovery Catalogue (GDC) must publish, every 24 hours, a zip file containing all metadata records available via the GDC API.

A GDC Sensor Centre checks all GDCs are publishing the zip file and then analyzes the content of the zip file (all metadata records) to detect anomalies (i.e. missing metadata records, inconsistent data records between GDCs, validating metadata records for WCMP2 compliance).

Metadata content must be identical for all GDCs. The SCGDC verifies this consistency provides the necessary insights to trigger corrective action of the discrepancies.

3.3.4.1 Technical considerations

- The SCGDC will connect to a minimum of two Global Brokers.
- The SCGDC subscribes to the metadata topic for all GDCs, by one of the following methods:
  - subscribing to `cache/a/wis2/+/metadata`, discarding all Notification Messages whose centre identifier (in the topic) does not end with `-global-discovery-catalogue`.
  - subscribing to each GDC via `cache/a/wis2/centre-id/metadata`.
- The SCGDC will use the link provided in the WIS2 Notification Message received to download the zip file published by the GDC.
- For each GDC centre identifier, upon successful download of the zip file, the SCGDC will update the metric `wmo_wis2_scgdc_last_download_timestamp_seconds` with the timestamp (in seconds) of the last successful download.
- Once a day, the SCGDC will analyze the content of the latest zip files received across all GDCs
  - Files older than 24 hours will not be analyzed.
- The SCGDC will set the metric `wmo_wis2_scgdc_records_incorrect_total` to `0` at the beginning of the analysis for all GDCs where the zip file is less than 24 hours old. **Note**: the metric `wmo_wis2_scgdc_records_incorrect_total` is a gauge (not a counter) whose value can be reset to `0` at the beginning of the analysis.
- After unpacking the zip file, the SCGDC will update the metric `wmo_wis2_scgdc_records_received_total` with the number of records found in the zip file.
- For each record, the SCGDC will:
  - prune the `links` array. The `links` array is modified by each GDC and modifications are dependent on the GDC itself. It is therefore required to remove all links that are not `rel=license` before further processing for parity checking.

- validate the record against the latest approved version of WCMP2
    - If the record is not compliant, the SCGDC will update the metric `wmo_wis2_scgdc_record_invalid_total`, with the WCMP2 id as well as centre id of the GDC
- compare each metadata record with the same WCMP2 `id` received from all other GDCs.
- If the content is the same for all GDCs, the SCGDC will increase the metric `wmo_wis2_scgdc_records_identical_total` by `1`.
- If the content is different, the SCGDC will compare the record's `properties.updated` property, if present, across GDCs. If more than one value of `properties.updated` is present, the SCGDC will consider the GDC with the record of the most recent date is the canonical record.
    - In this case, the SCGDC will increase the metric `wmo_wis2_scgdc_records_incorrect_total` by `1` for all GDCs except the GDC having the most recent record.
- If the content is different, and if no GDC holds a record with `properties.updated`, the SCGDC will consider that it is not possible to ensure which GDC has the correct record.
    - In this case, the SCGDC will increase the metric `wmo_wis2_scgdc_records_incorrect_total` by `1` for all GDCs.

- As a convention, the Sensor Centre for Global Discovery Catalogue centre-id will be `{tld}-{centre-name}-sensor-centre-global-discovery-catalogue`.

[1] https://docs.ogc.org/is/17-069r4/17-069r4.html

[2] https://docs.ogc.org/is/18-062r2/18-062r2.html

PART IV. Information management

4.1 Introduction

4.1.1 Background

The efficient and effective provision of services relying on meteorological, climatological, hydrological and oceanographic information depends on a reliable information infrastructure. This infrastructure should be guided by community best practices and standards, including recommendations and requirements on sourcing, securing, managing, archiving, exchanging and providing easy access to information. This part of the Guide aims to provide high-level guidance on these activities, which can be grouped under the term “information management”. This is done by identifying and describing the fundamental principles of good information management and by highlighting the different stages of the information management life cycle.

Note: The term "information" is used in a general sense and includes data and products.

4.1.2 Scope

High-level guidance on information management practices that apply in the context of information related to the Earth system is provided in this part of the Guide. Detailed technical information, such as the specification of data formats or quality control and assurance methods, is provided in other parts of the Guide and in other WMO publications. These are referenced where applicable.

The principles of information management are described below. Section [_3_3_the_information_management_life_cycle] describes five focus areas.

1. Planning, information creation and acquisition. The creation of information using internal and external data sources and the acquisition of information from various sources.
2. Representation and metadata. The use of standards to represent metadata, data and information; this is of primary importance to enable the interoperability and long-term usability of the information.
3. Publication and exchange of information. The creation and publication of discovery metadata in a standardized format, enabling users to discover, access and retrieve the information.
4. Usage and communication. The publication of guidance material on the use of published information, including on the limitations and suitability of the information and any licensing terms.
5. Storage, archival and disposal. The policies and procedures for business continuity and disaster recovery, as well as retention and disposal.

4.1.3 Intended audience

This guidance is primarily aimed at personnel within WMO centres, who are responsible for planning and undertaking the creation or acquisition, stewardship, exchange and provision of information related to the Earth system.

Specifically, the guidance has five main target audiences across the information life cycle:

1. Information producers or creators (those who produce or acquire the information) - they need to ensure the scientific quality of the underpinning information;
2. Information managers (those who manage the information);
3. Information providers or publishers (those who publish the information) - they are responsible for the provision of the information and for ensuring that appropriate access is enabled, licensing agreements are in place, and so forth;
4. Service providers (those who disseminate the information) - they are responsible for ensuring information availability and maintaining capability for easy and secure access to the information;
5. Information consumers (those who utilize the information) - they need to understand the restrictions, rights, responsibilities and limitations associated with the information together with the suitability for the intended usage or purpose.

4.2 Principles of information management

The effective management of information is essential for WMO centres to deliver operational services and information that is authoritative, seamless, secure and timely. The principles below underpin and provide a framework for information management across the full information life cycle. These principles are independent of the information type and are largely independent of technology; they are therefore expected to remain stable over time.

4.2.1 Principle 1: Information is a valued asset

- An information asset is information that has value. This value may be related to the cost of generating and collecting the information, it may be associated with the immediate use of the information, or it may be associated with the longer -term preservation and subsequent reuse of the information.
- This value should be recognizable and quantifiable and the asset should have an identifiable life cycle. Risks associated with, and to, an information asset should also be identified. As such, information management must be considered an integral part of a WMO centre's responsibilities and needs to be adequately resourced over the full life cycle of the information.

4.2.2 Principle 2: Information must be managed

- An information asset must be managed throughout its life cycle, from creation to use to eventual disposal, in a way that makes it valuable, maximizes its benefits and reflects its value in time and its different uses.
- Information managers must consider the entire information life cycle, from identifying needs and business cases to creation, quality assurance, maintenance, reuse, archival, and disposal. Careful consideration must be given to disposal, ensuring that information is destroyed only when it has ceased to be useful for all categories of users.
- Professionally qualified and adequately skilled staff with clear roles and responsibilities should apply a sound custodianship framework concerning security, confidentiality and other statutory requirements of different types of information.

4.2.3 Principle 3: Information must be fit for purpose

- Information should be developed and managed in accordance with its function and use for internal and external users.
- WMO centres should regularly assess information to ensure that it is fit for its purpose and that the related processes, procedures and documentation are adequate.
- Processes should be consistent with the general provisions and principles of quality management as described in the WMO *Technical Regulations* (WMO-No. 49).

4.2.4 Principle 4: Information must be standardized and interoperable

- Information must be stored and exchanged in standardized formats to ensure wide usability in the short and long-term. It is essential for long-term archiving that information be stored in a form that can be understood and used after several decades.
- Standardization is essential for structured information such as dataset definitions and metadata to support interoperability.
- Interoperability is essential for users to be able to utilize information through different systems and software. Open standards help ensure interoperability with their openness and wide adoption across various communities.
- Which standards to use depends on the user community and organizational policies. Interoperability requirements should be considered when selecting the standard for internal use and broader dissemination.
- The use of closed and proprietary standards is strongly discouraged.

4.2.5 Principle 5: Information must be well documented

- WMO centres should comprehensively document information processes, policies, and procedures to facilitate broad and long-term use.
- WMO centres should keep documentation up to date to ensure full traceability of processes along the information life cycle, particularly for its creation.
- Previous versions of the documentation should be retained, versioned, archived and made readily available for future use. In addition, versions should be assigned a unique and persistent identifier for future unambiguous identification.

4.2.6 Principle 6: Information must be discoverable, accessible and retrievable

- Information should be easy to find through the Web, and for this purpose, the publisher should share discovery metadata with a catalogue service. The catalogue service should include a web API to be used by other applications in order to offer user-tailored search portals.
- For information to be easily retrievable once discovered, it should be accessible using standard data exchange protocols.

4.2.7 Principle 7: Information should be reusable

- In order to maximize the economic benefits of an information asset, it should be made as widely available and as accessible as possible.

- Resolution 1 (Cg-Ext(2021)) encourages the reuse of data and information through the open and unrestricted exchange of core WMO data. WMO encourages the free and unrestricted exchange of information in all circumstances.
- The publisher should provide an explicit and well-defined licence for each information item or dataset as part of the associated metadata.
- The Findable, Accessible, Interoperable and Reusable (FAIR) data principles promote open data with the ultimate goal of optimizing the reuse of data. These principles should be followed where possible.

Note: Information on the FAIR data principles can be found at: FAIR Principles - GO FAIR[1]

4.2.8 Principle 8: Information management is subject to accountability and governance

- Information management processes must be governed as the information moves through its life cycle. All information must have a designated owner, steward, curator and custodian. These roles may be invested in the same person but should be clearly defined at the time of creation. A WMO centre with responsibility of managing information must ensure:
  - The implementation of general information management practices, procedures and protocols, including well-defined roles, responsibilities and restrictions on managing the information;
  - The definition and enforcement of an appropriate retention policy, taking into account stakeholder needs and variations in value over the information life cycle;
  - The establishment of licensing and the definition and enforcement of any access restrictions.
  - The designated owner should have budget and decision-making authority with respect to preservation and data usage, including the authority to pass ownership to another entity.

4.3 The information management life cycle

4.3.1 Overview

All information should be subject to a well -defined and documented life cycle. The governance of this process is often referred to as the information management life cycle; it helps organizations manage information from planning, creation and acquisition through usage and exchange to archival and disposal.

The following sections describe two overarching themes, governance and documentation, which apply to all stages of the information life cycle; these sections provide high -level guidance and are split into five aspects:

- Planning, creation and acquisition;
- Representation and metadata;
- Publication and exchange;
- Usage and communication;
- Storage, archival and disposal.

Governance covers the rules that apply to managing information in a secure and transparent manner; documentation covers the act of recording the reasons for, and details of, all operations in the information management process.

4.3.2 Overarching requirements

4.3.2.1 Governance

- Information management governance defines a set of organizational procedures, policies and processes for the management of information. This includes defining accountabilities and compliance mechanisms.
- Effective governance helps ensure that all aspects of the information management process are conducted in a rigorous, standardized and transparent manner and that the information is secure, accessible and usable.
- WMO centres should establish a board or leadership group to develop and regularly review such a governance structure and ensure compliance with its requirements.

4.3.2.2 Documentation

- Documentation describing the who, what, why, when, where and how with respect to the various actions that are undertaken in the management of information is required to ensure the traceability and integrity of the information and to ensure operations can continue if key staff leave.
- This documentation is required for all aspects of the information life cycle and should be clear, well -communicated, regularly updated and easy to find. Guidance relating to the documentation should be provided to new staff taking on responsibilities for information management and be a key component of training.

4.3.3 Aspects of the information management life cycle

4.3.3.1 Planning, information creation and acquisition

Before the creation or acquisition of new information a business case plan and an information management plan should be developed, covering both the input information sources and any derived information. The plans should include:

- Why the information is required;
- How it will be collected or created;
- How it will be stored;
- Whether it will be exchanged with other users and under what policy;
- Where it should be submitted for long-term archival;
- Key roles and responsibilities associated with the management of the information.

For externally sourced data, the plans should include where the information has come from and what the licensing terms are.

Once information has been acquired, it should be checked to ensure that the contents and format are as expected. This may be done using a compliance checker or a validation service. Once these checks have been performed, the information content should also undergo quality control checks using well-documented procedures to identify any issues. A record of the checks should be kept, and any issues detected should be documented and sent back to the originators. It is also important to subscribe to updates from originators so any issues identified externally can be taken into account.

Information created rather than acquired should undergo the same processes as acquired information. Information created should undergo quality control, and the resulting files should be checked against the specified format requirements. The results of the processes and checks should be documented.

To ensure traceability and reproducibility, the information and documents at this and subsequent stages, should be version controlled and clearly labelled with version information. Similarly, software or computer code used to generate or process information should be version controlled with the version information recorded in the documentation and metadata. Where possible, software should be maintained within a code repository.

4.3.3.2 Representation and metadata

The formats used to store and exchange information should be standardized to ensure its usability in both the short and the long-term. It is essential that the information be accessible many years after archival if required. To ensure this usability, the format and version of the information should be recorded in the information metadata record and included within the information itself where the format allows.

Information exchanged on WIS and between WMO centres is standardized through the use of the formats specified in the *Manual on Codes* (WMO-No. 306), Volume I.2 and the *Manual on WIS*, Volume II. These include the GRIB and BUFR formats for numerical weather prediction products and observational data and the WMO Core Metadata Profile for discovery, access and retrieval metadata. The format for the exchange of station and instrumental metadata, WMO Integrated Global Observing System (WIGOS) Metadata Data Representation, is defined in the *Manual on Codes* (WMO-No. 306), Volume I.3.

These formats have been developed within the WMO community to enable the efficient exchange of information between WMO centres and to enable the information to be interoperable between centres and systems. The formats, including detailed technical information, have also been published in WMO manuals, permitting other communities to use the formats and the information and promoting the reuse of the information.

The WMO formats specified in the manuals are subject to strong governance processes, and changes to the formats can be traced through the versions of the manuals. The code tables and controlled vocabularies are also maintained in a code repository. To enable future reuse, the technical information, including detailed format specifications, should be archived alongside information for future access. This includes any controlled vocabulary, such as BUFR tables or WIGOS metadata code lists, associated with the format.

4.3.3.3 Publication and exchange of information

To maximize the benefits and return on investment in the acquisition and generation of information, there needs to be a clear method as to how the information will be published, exchanged and accessed by users.

Information is published on WIS through the creation of discovery metadata records. These records are publicly searchable and retrievable via WMO cataloguing services, providing access to the records via the Web and via a web API. The metadata records should include information on how to access the described datasets and services (see *Manual on WIS*, Volume II – Appendix F. WMO Core Metadata Profile (Version 2)) and how to subscribe to receive updates and new data.

Technical regulations are provided in the *Manual on WIS*, Volume II. Before exchange and publication, the metadata should be assessed using the WMO Core Metadata Profile KPIs to ensure usable and high -quality metadata in addition to metadata that conform to the technical standard.

The web standards and protocols used should be adequately documented to enable users to find and retrieve the information. This should be possible both manually and automatically via machine-to-machine interfaces and should be standardized between centres.

Updates to the information exchanged on WIS, including the publication of new information or the cessation of previously exchanged information, are published in the WMO Operational Newsletter.

Note: The newsletter is available from: https://community.wmo.int/news/operational-newsletter.

4.3.3.4 Usage and communication

For information to have value, it must inform users, aid knowledge discovery and have an impact through informed decision -making. Ensuring that the user can make effective use of the information is an important step in the information management life cycle. This is accomplished in two ways:

1. By providing suitable information within the discovery metadata, enabling users to discover and access the information, including licensing information, and to assess whether it meets their requirements;
2. By providing user guides and documentation on the suitability of the information for different uses, including any technical caveats or restrictions on the use of the information.

For common types of information, the guides may be generic or link to standard documentation. Information on the observations available from WIGOS is provided in the *Manual on the WMO Integrated Global Observing System* (WMO-No. 1160) and the *Guide to the WMO Integrated Global Observing System* (WMO-No. 1165). This includes information on the expected uses and quality of the data. Similarly, information on the data and products available through the WMO Integrated Processing and Prediction System is provided in the *Manual on the WMO Integrated Processing and Prediction System* (formerly the Manual on the Global Data Processing and Forecasting System) (WMO-No. 485).

For non-standard and specialist products, targeted user guides may be more appropriate. These should be accessible and retrievable via a link within the discovery metadata and should include a plain text summary for the non-technical user. Any user guide should be in addition to the technical documentation described in [_3_3_3_1_planning_information_creation_and_acquisition].

Updates and the availability of new information should be announced and published via the WMO

Operational Newsletter (see [_3_3_3_3_publication_and_exchange_of_information]). Other communication methods may also be used, but these should not be in place of the operational newsletter. It is also recommended that users be allowed to subscribe to the newsletter to receive updates directly.

The discovery metadata should include a valid point of contact, enabling users to provide feedback and ask questions about the information provided.

4.3.3.5 Storage, archival and disposal

The type of storage used should be appropriate to the type of information stored. Core information exchanged operationally should be stored and made available via high-availability and low-latency media and services. For some operation-critical information, such as hazard warnings, there is a requirement for the end-to-end global distribution of the information to be completed in two minutes. For other operational data, there is a requirement for the global exchange to be completed in 15 minutes.

The storage requirements for non-operational services and information may be different, but the guidance provided in this section applies equally. Further information on the performance requirements is provided within the WIS2 technical specifications listed in the *Manual on WIS*, Volume II.

Backup policies and data recovery plans should be documented as part of the information management plan. They should be implemented either before or when the information is created or acquired and should include both the information and the associated metadata. The backup and recovery process should be routinely tested.

Business rules governing access to and modification of the information should be clearly documented in the information management plan. These must include the clear specification of the roles and responsibilities of those managing the information. Information on who can authorize the archival and disposal of the information and the processes for doing so should be included. The roles associated with an information resource are standardized as part of the WMO Core Metadata Profile.

The archival and long-term preservation of an information resource should be identified and included in the information management plan. It may take place at a national data centre and/or a WMO centre. WMO centres are recommended for globally exchanged core data and include those centres contributing to the Global Atmosphere Watch, the Global Climate Observing System and Marine Climate Data System (see *Manual on Marine Meteorological Services* (WMO-No. 558), Volume I, as well as the WMO World Data Centres and in the *Manual on WIS*, Volume II and those defined in the *Manual on the WMO Integrated Processing and Prediction System* (formerly the *Manual on the Global Data Processing and Forecasting System*) (WMO-No. 485).

Earth system information, especially observational data, is often irreplaceable. Other information, while technically replaceable, is often costly to produce and therefore not easily replaceable. This includes outputs from numerical models and simulations. Before an information resource is marked for disposal, careful consideration must be given to whether long-term archival or disposal is more appropriate. This consideration must follow a clearly defined process documented in the information management plan.

When an information resource is marked for disposal, the reasons for disposal, including the outcome of the consultation with stakeholders and users, must clearly be documented. The disposal must be authorized by the identified owner and custodian of the information. Information relating to the disposal must be included in the metadata associated with the information resource. The metadata must be retained for future reference.

4.4 Other considerations

4.4.1 Technology and technology migration

Information managers must be aware of the need to ensure that the technologies, hardware and software used do not become obsolete, and they must be aware of emerging data issues. This topic is discussed further in the *WMO Guidelines on Emerging Data Issues* (WMO-No. 1239).

4.4.2 Information security

Further information on information security and best practices can be found in the *Guide to Information Technology Security* (WMO-No. 1115).

[1] https://go-fair.org

PART V. Security

For this initial version of the Guide to WIS, Volume II, existing guidance on information technology security (also known as "cybersecurity") remains largely applicable. Please refer to:

- *Guide to Information Technology Security* (WMO-No. 1115);
- *Guide to the WMO Information System* (WMO-No. 1061), Volume I, Appendix E. Annex To Paragraph 7.8, 1. ICT Service Incident Management; and Appendix F.WIS IT Security Incident Response Process.

Annex A: Revision History

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PART VI. Competencies

For this initial version of the *Guide to WIS*, Volume II, existing guidance on competencies remains largely applicable. Please refer to *Guide to the WMO Information System* (WMO-No. 1061), Volume I, Appendix A. WMO Information System Training and Learning Guide.